

Open Science Research Integrity Toolkit

27 empirical techniques for advancing research integrity – organized by the level of the problem (tier) and issue type

TIER 1

Individual-Level Misconduct & Errors

Tools that examine a single paper's statistics, images, or text for signs of fabrication, manipulation, or error — without requiring access to raw data.

Data Fabrication

A researcher invents or alters data. These tools detect impossible or implausible summary statistics in published papers.

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| 01 | GRIM Test | Tests whether a reported mean is possible given sample size and integer scale. Requires only the reported mean and N — no raw data. |
| 02 | SPRITE | Reconstructs all integer datasets consistent with reported M, SD, and N. Exposes physically impossible or implausible data distributions. |

Image Manipulation

Figures are duplicated, spliced, or AI-generated. Tools identify pixel-level duplication patterns within or across manuscripts.

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| 03 | Bik Visual Screening | Manual comparison of published figures for identical regions passed off as distinct experiments. Requires full-resolution PDF or image files. |
| 04 | Imagetwin AI | AI-powered batch scanning for pixel-level image duplication. Deployed by publishers at submission. Requires good-quality PDF input. |
| 05 | Proofig AI | Publisher-grade image integrity checking before peer review. Detects duplicated panels, background uniformity, and contrast manipulation. |

Plagiarism & Text Recycling

Text is copied or AI-generated without attribution. Detection is harder as paraphraser preserve meaning while evading exact-match tools.

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| 06 | iThenticate / CrossCheck | Text similarity matching against published papers, preprints, and web content. Cannot detect meaning-preserving paraphrase. |
| 07 | Tortured Phrases Screener | Flags machine-translation obfuscation used to evade plagiarism detectors — signature of paper-mill produced manuscripts. |
| 08 | AI-Text Detectors | Identifies probabilistic signatures of LLM-generated text. Reliability is limited after deliberate paraphrasing; use as a screening flag only. |

Research Methods Errors

Coding bugs and data-entry errors that are unintentional but consequential. Tools reconstruct implied computations and flag implausible values.

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| 09 | statcheck | Extracts t, F, χ^2 , z, and r statistics from a PDF and recalculates p-values. Flags decision-changing mismatches. Requires machine-readable PDF. |
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TIER 2a

Collaborative Pipeline Failure

Errors emerging from handoffs between team members on interdependent code and data. One upstream change cascades silently through all downstream components.

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| 10 | Registered vs. Published Outcome Audit | Compares pre-registered hypotheses and outcomes to those in the final paper. Divergences signal HARKing, outcome switching, or pipeline changes. |
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Team-Level Analytic Issues

Tools that detect problems arising from collective analytic decisions — p-hacking, selective reporting, model fragility — applied across a set of studies or specifications.

p-Hacking / Selective Reporting

Teams run many analyses and report only significant results. Tools examine p-value distributions across a literature to detect selective reporting signatures.

11	p-curve	Plots distribution of significant p-values. Genuine effects → right-skewed. p-hacking → flat or left-skewed. Requires test statistics and df per finding.
12	p-uniform	Models full p-value distribution under a specific assumed effect to estimate the true effect size and test publication bias. Requires effect sizes and p-values.
13	Caliper Test	Tests for excess of results just below $p < .05$ — a signature of selective stopping or rounding. Requires p-values from many studies in the same literature.
14	z-curve 2.0	Fits a mixture model to all z-scores (significant and non-significant). Estimates expected replication rate and discovery rate. Requires 100+ test statistics.

Fragile Findings & Analytical Fragility

Results that depend on undisclosed analytic choices. Tools run all defensible specifications to reveal whether significance is robust or narrow.

The Garden of Forking Paths

Three tools for mapping how much a finding depends on the particular choices made from a larger space of equally defensible analytic options.

15	Multiverse Analysis	Runs every defensible combination of analytic choices and plots the distribution of resulting effect sizes. Requires access to raw data.
16	Specification Curve Analysis	Adds inferential permutation tests to the multiverse approach — asks whether observed results exceed a null baseline. Requires raw data.
17	Vibration of Effects	Cycles through all subsets of candidate covariates; records effect estimate for each. VoE ratio > 2 or sign changes indicate fragility. Requires individual-level data.

Publication Bias (Team & PI Decisions)

Teams selectively submit significant results. Tools detect the statistical signature of missing studies and estimate inflation of the published effect size.

Correcting for the File Drawer

Four meta-analytic tools that detect and correct for publication bias — the systematic suppression of null results from the published record.

18	Funnel Plot & Egger Test	Plots effect size vs. precision. Asymmetry signals missing null studies. Egger test formalizes this as a regression. Requires ≥ 10 studies, effect sizes and SEs.
19	Trim-and-Fill	Imputes missing studies to restore funnel symmetry; re-estimates the bias-corrected pooled effect. Assumes missing studies mirror observed asymmetric ones.
20	PET-PEESE	Meta-regression: regresses effect size on SE (PET) then SE^2 (PEESE) to estimate the true effect after removing bias. Best with ≥ 20 –30 studies.
21	Excess Significance Test (TES)	Compares observed to expected number of significant results given study power. Large excess implies p-hacking, publication bias, or both.

TIER 2b

Group / Collusion-Level Misconduct

Tools that detect coordinated misconduct across multiple authors, institutions, or journals — analyzing relationships between entities rather than individual paper content.

Citation Rings & Uncritical Peer Review

Dense mutual-citation or mutual-review networks inconsistent with field norms. Detection requires bibliometric analysis of relational patterns.

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| 22 | Citation Network Analysis | Maps citation relationships to identify anomalously dense clusters. Uses network centrality and modularity. Requires access to bibliometric databases. |
| 23 | Bibliometric Anomaly Detection | Flags journals or author groups with anomalous self-citation or co-author citation rates compared to field baselines. Requires citation metadata. |

Fake Peer Review Rings

Authors submit fabricated reviewer emails or collude with editors for pre-agreed reviews. Now accounts for roughly half of all retractions globally.

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| 24 | Editor Network Analysis | Maps editorial submission routing to identify editors disproportionately handling each other's manuscripts. Requires publisher submission metadata. |
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Paper Mills & Predatory Journals

Industrial fabrication networks producing manuscripts for purchase. Detection uses text-level obfuscation signatures and image clustering across large paper sets.

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| 25 | Problematic Paper Screener | Scans papers for tortured phrases, image duplication, and metadata anomalies. Has flagged 32,786+ suspected paper-mill products. Works from PDF and metadata. |
| 26 | Hijacked Journal Checker | Database of 400+ journal ISSNs cloned by scam publishers mimicking legitimate journals. Functions as a lookup tool — requires only journal name or ISSN. |

TIER 3

Field / Community-Level Problems

Tools that assess the integrity or reliability of an entire research literature — quantifying systemic biases in how knowledge is produced, reported, and preserved.

Poor Data Stewardship & Inaccessible Research

Data and code are lost, unavailable despite stated commitments, or deposited in unusable formats. Tools assess whether research meets minimum reproducibility standards.

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| 27 | FAIR Compliance Assessment Tools | Automated scoring (FAIRshake, F-UJI) of whether a dataset meets the FAIR principles: Findable, Accessible, Interoperable, Reusable. Requires only the repository landing page. |
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THE GRIM TEST

Granularity-Related
Inconsistency of
Means — detecting
impossible reported
averages in studies
with integer response
scales

At a glance: The GRIM test checks whether a reported mean is arithmetically possible given the sample size and the fact that the underlying data are integers. If no combination of integer responses could produce the reported mean, the paper contains an error — or something worse.

1. Overview

When survey or experimental data are collected on integer response scales — a 7-point Likert scale, a count of correct answers, a yes/no score converted to 0/1 — every data point is a whole number. The mean of a set of integers is constrained: not every decimal value is reachable given the number of observations. The GRIM test exploits this constraint.

Introduced by Nicholas Brown and James Heathers in 2016, the test is simple enough to run with pencil and paper or a spreadsheet, but it proved immediately powerful: when applied to papers in leading social psychology journals, they found inconsistencies in a substantial fraction of tested articles — most of which had passed peer review.

The test does not prove fabrication. An impossible mean could arise from a transcription error, a rounding mistake, or a copy-paste problem. But it proves that something in the paper is wrong — itself significant when the paper has influenced downstream work or clinical guidelines.

2. Methodological Overview

The core intuition is easiest to grasp with a small example. Suppose ten participants each rate their agreement on a scale from 1 to 7. The sum of their responses must be an integer between 10 and 70. The mean is that integer sum divided by 10 — so the mean can only take specific decimal values such that (mean $\times$ N) is a whole number.

More precisely: multiply the reported mean by the reported N. If the result is not (very close to) a whole number, the mean is impossible. The test formalises this into a single check:

$$(\text{Reported Mean} \times N) \bmod 1 \approx 0$$

Where mod 1 gives the fractional part. If it is not negligibly close to zero (or to one, to allow for rounding), the mean cannot exist for integer data with that sample size.

In practice, you need to account for rounding: a reported mean of 3.47 could represent a true mean anywhere between 3.465 and 3.475. The test checks whether any integer sum within that rounding window is achievable. If no integer multiple falls within the window, the mean is impossible regardless of rounding. If one does, the mean is GRIM-consistent.

The test extends naturally to standard deviations (GRIMMER, Allard 2018) and to subgroup means, percentages, and proportions. Each extension applies the same core logic: what integer sums are possible given N, and does the reported statistic correspond to one of them?

3. Example Application: Wansink Food Lab

Brian Wansink's Cornell Food and Brand Lab produced dozens of widely-cited findings about how environmental cues affect eating behavior. Following concerns raised in 2016, James Heathers and Nicholas Brown applied GRIM systematically to the lab's published papers.

One finding: a paper reported that children who saw an Elmo sticker on an apple chose the apple 71.7% of the time in a sample of N = 23. Multiply: $0.717 \times 23 = 16.491$. This is not a whole number — no arrangement of 23 children can produce a mean of 0.717. The reported percentage is impossible.

Across the Wansink papers, Heathers and Brown identified dozens of GRIM-inconsistent means. When SPRITE was applied alongside GRIM, the pattern was too systematic to be explained by transcription error — including one finding implying a child ate approximately 60 carrots at a single sitting.

Cornell University eventually retracted 13 of Wansink's papers following a formal misconduct investigation. The GRIM test was the statistical tripwire that initiated the scrutiny.

4. Requirements & Applicability

The GRIM test is not universally applicable. Its usefulness depends on properties of the data and what the paper actually reports.

Sample size (N)	Must be reported. Most powerful when $N < 60$ per group — with larger N , more decimal values become reachable and the test rarely fires.
Scale type	The underlying response scale must be integer-valued: Likert scales, binary 0/1 variables, counts, ordinal scores. Continuous measurements cannot be tested.
Decimal precision	The mean must be reported with at least one decimal place. A mean reported as an integer is GRIM-consistent with any N — the test only gains power when rounding is observable.
Reported statistic	Means or proportions are the primary target. Does not apply directly to SDs, correlations, or p-values — though GRIMMER and SPRITE extend the logic.
Raw data required?	No. The test operates entirely on published summary statistics — the reported mean and N . This is what makes it feasible as a post-publication check by anyone with access to the paper.
Scale bounds	Knowing the minimum and maximum response value allows an additional bounds check: the mean must fall within [scale min, scale max]. Important for proportion-based tests.

5. Limitations

- Cannot detect fabrication in continuous data (reaction times, physiological measurements). The majority of findings in many domains are not testable.
- A GRIM-consistent result is not validated — it simply cannot be ruled out. Many errors and most fabrications will not produce GRIM-inconsistent means.
- The test loses power as N grows. For $N > 100$, almost any decimal mean is achievable and inconsistencies are rarely flagged.
- Weighted means, trimmed means, or means computed after data transformation may appear GRIM-inconsistent without being wrong. Context matters.
- Requires knowing the response scale. Papers that do not report the scale, or that aggregate across multiple differently-scaled items, may produce apparent inconsistencies that are artefacts of aggregation.

6. How to Apply

Manual / spreadsheet method

- Identify every mean (or proportion) reported in the paper along with its associated N .
- Multiply each mean by its N to get the implied integer sum.
- Check whether the implied sum is very close to a whole number — compute $(\text{mean} \times N)$ and check whether the fractional part is < 0.001 or > 0.999 .
- Account for rounding: check whether any integer within the rounding window ($\text{mean} \pm 0.005$ for two-decimal reporting) produces a whole number sum when multiplied by N .
- Flag any means where no integer sum is reachable even after allowing for rounding.

Web-based tool

The GRIM app (grimapp.site, maintained by Heathers and Brown) accepts a mean, N, and scale bounds and returns an immediate consistency verdict with a visualization of which integer sums are reachable.

R / Python

The grimmer R package (Anaya 2016) implements both GRIM and GRIMMER for batch processing. It accepts a data frame of (mean, N, item_count, scale_min, scale_max) tuples and returns a consistency flag for each.

7. Related Tools

SPRITE	Extends GRIM: instead of asking 'is this mean possible?', SPRITE asks 'what distributions of integer responses could produce this mean and standard deviation together?' Reveals when the combination of M and SD implies a physically impossible or implausibly narrow data distribution.
statcheck	Automated extraction and recalculation of p-values from PDFs. Operates on test statistics rather than means; complements GRIM by catching a different class of reporting error.
GRIMMER	Extends GRIM to standard deviations: checks whether the reported SD is achievable given M, N, and integer constraints. Substantially more demanding mathematically but follows the same core logic.

Key Citations

- Brown, N. J. L. & Heathers, J. A. J. (2017). The GRIM test: A simple technique detects numerous anomalies in the reporting of results in psychology. *Social Psychological and Personality Science*, 8(4), 363–369. <https://doi.org/10.1177/1948550616673876>
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SPRITE

Tier 1 :: Data Fabrication

At a glance: SPRITE (Sample Parameter Reconstruction via Iterative Techniques) goes beyond asking whether a single reported mean is arithmetically possible. Given a reported mean, standard deviation, sample size, and response scale, SPRITE reconstructs the full space of integer datasets that could have produced those summary statistics. When no such dataset exists, or when every possible dataset implies physically implausible patterns — such as responses clustered impossibly at one end of a scale — the paper contains a fundamental inconsistency. Like GRIM, SPRITE requires only published summary statistics and no raw data.

1. Overview

The GRIM test answers a binary question: is this reported mean arithmetically possible given the sample size and integer scale? SPRITE asks a richer question: what do the actual response distributions look like that could have produced this combination of mean, standard deviation, and sample size? Rather than a pass/fail verdict, SPRITE returns a picture — a set of plausible datasets — and asks whether that picture is coherent with reality.

SPRITE was developed by James Heathers, Jordan Anaya, Tim van der Zee, and Nick Brown, and first circulated as a preprint in 2018. It extends the logic of GRIM in two important directions. First, it incorporates the standard deviation as well as the mean, which dramatically tightens the constraints on what datasets are possible: a mean that is achievable by many integer distributions may be achievable by only a handful when the SD is also constrained. Second, it can reveal that even though a mean-SD combination is technically possible, every dataset that could produce it is physically implausible — implying extreme distributions that no realistic sample of participants would generate.

The method has its most powerful application in studies using bounded integer scales — Likert-type rating instruments, binary outcomes, small-range counts — where the constraints on possible distributions are strongest. It is less informative for continuous measurements or very large samples, where the space of possible distributions becomes so large that the constraints lose their bite.

2. Methodological Overview

SPRITE works by computational enumeration. The algorithm generates a large number of random integer datasets of size N drawn from the reported scale range $[\min, \max]$, then iteratively adjusts pairs of values until both the target mean and target standard deviation are simultaneously achieved. Each valid dataset found is a member of the SPRITE solution set — a plausible reconstruction of what the underlying data could have looked like.

The iterative reconstruction algorithm

Starting from a random distribution with the correct mean (obtained by adjusting values until $\Sigma x/N$ matches the reported M), the algorithm applies the following iterative step:

1. Select two values at random: x_i and x_j from the current distribution.

The swap step: move x_i and x_j slightly toward or away from each other by 1 unit.

2. Accept the swap if it brings the current SD closer to the target SD.

Reject it if the swap moves the SD further away, or if it pushes x_i or x_j outside $[\min, \max]$.

3. Repeat until SD matches the target (within rounding tolerance).

Each successful run produces one member of the solution set. Run thousands of times to explore the full solution space.

The result is a large sample from the space of possible datasets. This sample can then be examined visually (as a histogram grid showing each reconstructed distribution) or analytically (checking whether the distributions cluster in plausible regions of the scale or exhibit impossible features such as bimodality with all responses at the extremes).

What SPRITE detects

Three types of anomaly are identifiable from the SPRITE solution set:

- Empty solution set: no integer dataset of size N can simultaneously achieve the reported M and SD within rounding tolerance. This is a direct impossibility — the summary statistics are internally inconsistent, regardless of what the underlying data looked like. This is a stronger finding than a GRIM failure alone.
- Implausible distribution shapes: the solution set exists but every distribution in it has an extreme or physically implausible shape. A classic example is a distribution where all N participants must have responded at either the minimum or maximum of the scale, with no intermediate responses — a pattern inconsistent with normal human response behavior on a rating scale. Or a distribution where more than half of participants gave identical responses, when the study design makes such clustering highly improbable.
- Inconsistency across subgroups: a paper reports means and SDs for multiple subgroups, each of which passes SPRITE individually. But when the subgroup distributions are combined, the resulting pooled distribution is impossible given the reported overall mean or SD. SPRITE can detect this cross-subgroup inconsistency by simultaneously constraining multiple distributions.

3. SPRITE vs. GRIM: Complementary Constraints

GRIM and SPRITE operate on the same class of data — integer-scale responses with reported summary statistics — but they constrain different aspects of the summary and therefore catch different classes of error. Understanding the relationship helps clarify when each is more useful.

What it tests	GRIM: Is the reported mean achievable for any integer dataset of this N? SPRITE: What distributions could produce this exact M and SD combination?
Information used	GRIM: Mean and N only. SPRITE: Mean, SD, N, scale minimum, and scale maximum — all five.
Output	GRIM: Binary pass/fail verdict. SPRITE: A set of plausible distributions, which may be empty, implausible, or normal.
Error types detected	GRIM: Impossible means (fabricated or transcription error). SPRITE: Impossible M-SD combinations; implausible distributions; cross-subgroup inconsistencies.
Power vs. N	GRIM: Loses power as N grows (many decimal means become achievable). SPRITE: Also weakens at large N, but SD constraint keeps it informative for moderate N (up to ~100).
Computational cost	GRIM: Instant — single arithmetic check. SPRITE: Requires iterative search; computationally heavier but still fast (seconds per statistic in R).

In practice, SPRITE is applied after GRIM as a second-pass analysis. GRIM rules out impossible means quickly; SPRITE then interrogates the plausibility of the underlying distribution for the statistics that pass the GRIM check but still merit scrutiny.

4. Example Application: Wansink Food Lab

The most extensively documented application of SPRITE is to Brian Wansink's Cornell Food and Brand Lab papers, the same set that prompted the original GRIM analysis by Brown and Heathers. After GRIM identified dozens of impossible means across the lab's published papers, Heathers and colleagues applied SPRITE to the mean-SD pairs that had survived the GRIM screen.

One set of findings involved a study on how plate size affects the amount of food served. The paper reported that children given large plates served themselves a mean of 59.8 grams of pasta with an SD of 47.3, N = 43. GRIM passes this ($59.8 \times 43 = 2571.4$ — not quite an integer, but within rounding tolerance for one decimal place). SPRITE, however, revealed that every distribution capable of producing $M = 59.8$, $SD = 47.3$, $N = 43$ on a plausible gram-range scale involved an implausible concentration of values at very low quantities (near zero grams) combined with a few extreme outliers at very high quantities. The shape was inconsistent with a normal serving-size distribution from a real cafeteria study.

A second example: a study reported a mean of 2.50, SD = 0.54, N = 20 on a 5-point scale. SPRITE found that the only distributions satisfying these constraints had exactly 20 participants responding exclusively with 2s and 3s — a distribution so perfectly symmetric and limited in range that it is extraordinarily unlikely to occur in real behavioral data, where participants typically show more variability.

These analyses, combined with the GRIM findings, contributed to the broader pattern of evidence that led Cornell University to investigate the lab. The investigation ultimately concluded that data had been manipulated, and 13 papers were retracted.

5. Requirements & Applicability

SPRITE shares the core requirement of GRIM — integer-scale response data with reported summary statistics — but adds the standard deviation as an essential input. This means it is applicable to a narrower set of reported statistics than GRIM, but each application is more powerful.

Mean (M)	The reported mean to four significant figures. More decimal places provide stronger constraints. A mean rounded to one decimal place leaves a wider solution window than a mean reported to two or three.
Standard deviation (SD)	Required — this is what distinguishes SPRITE from GRIM. The SD must be reported with at least one decimal place. If only standard errors (SE) are reported, these must be converted: $SD = SE \times \sqrt{N}$.
Sample size (N)	Required. SPRITE's constraints tighten as N decreases; the method is most powerful for $N < 100$, and the solution space becomes very large for N above ~200.
Scale minimum and maximum	Required. SPRITE generates integer values within [min, max]. An incorrect scale range (e.g., treating a 0–6 scale as a 1–7 scale) will generate a misleading solution set. Scale bounds must be verified from the paper's methods section.
Integer scale	The underlying data must be integer-valued. Continuous measurements, log-transformed variables, or composite scores computed from multiple items cannot be tested directly. Individual item responses on a Likert scale are SPRITE-applicable; scale means aggregated across items are not (their values are not integers).
Raw data required?	No. Like GRIM, SPRITE operates entirely on published summary statistics. This is the key advantage: it can be applied by any reader with access to the paper, without requiring data sharing from the original authors.
Number of statistics	SPRITE is typically applied to a set of statistics from the same paper, not just a single M-SD pair. Patterns of anomaly across multiple statistics in the same paper are more diagnostic than any individual result.

6. Limitations

- SPRITE loses power rapidly as N increases. For N above ~200, the space of possible integer distributions is so large that almost any M-SD combination is achievable, and the solution set rarely has a constrained shape. The method is primarily useful for small-to-medium sample studies ($N < 100$).
- The method requires correct scale bounds. If an analyst uses the wrong scale range — for example, treating a 7-point scale anchored at 0 as running from 1 to 7 — the solution set will be miscalculated. Always verify scale range from the methods section before running SPRITE.
- SPRITE identifies implausible distributions, but implausibility is a judgment, not a bright line. A solution set containing only extreme bimodal distributions is highly suspicious; a solution set with mild skew is less so. Interpreting SPRITE output requires domain knowledge about what response distributions look like for the specific construct and population being studied.
- Rounding cascades can produce false positives. If means and SDs were both rounded independently before reporting, the combination may appear to violate SPRITE constraints even when the underlying data were entirely legitimate. More decimal places in reported statistics reduce this risk. When a SPRITE solution set is empty by a tiny margin, rounding error is a plausible explanation.
- SPRITE cannot determine the cause of an anomaly. An empty solution set or implausible distribution could reflect fabrication, data manipulation, copy-paste errors, incorrect scale reporting, or a genuinely

unusual data distribution. SPRITE flags the inconsistency; establishing its cause requires additional investigation.

- The iterative algorithm is stochastic. Different runs may produce slightly different solution sets, especially for distributions near the boundary of feasibility. Running SPRITE multiple times (the rSPRITE R implementation handles this automatically) and examining the aggregate solution space is better practice than relying on a single run.

7. How to Apply

rSPRITE (R package)

The rSPRITE package (Brown & Heathers, maintained on GitHub) is the standard implementation. It accepts a data frame of (mean, sd, n, scale_min, scale_max) rows and for each row generates a set of reconstructed distributions, plots them as histograms, and flags cases where the solution set is empty or all solutions are implausible.

Minimal workflow

- Install the package: `devtools::install_github('nicebread/rSPRITE')` in R.
- Build the input: create a data frame with columns mean, sd, n, min_val, max_val for each M-SD pair to test. Verify scale bounds from the paper's methods section before entering them.
- Run the analysis: `sprite_results <- rsprite2(df)`. The function returns a list of solution sets, one per row. For each, inspect the histogram grid of reconstructed distributions.
- Flag anomalies: cases where `solution_count == 0` are impossible. Cases where all solutions show extreme concentration at scale endpoints or implausibly low within-distribution variance warrant documentation and further scrutiny.
- Apply across multiple statistics: SPRITE is most informative when run on all reported M-SD pairs in a paper simultaneously. Patterns of anomaly — multiple consecutive failures, or all solutions in multiple analyses being implausible — provide stronger evidence than any individual result.

Online tool

A web implementation is available at the SPRITE online app maintained by Brown and Heathers (searchable via GitHub). It accepts individual M, SD, N, min, max entries and returns a visual plot of the solution space without requiring R installation — useful for quick screening of individual statistics before committing to a full batch analysis.

8. Related Tools

GRIM Test	The natural prerequisite: GRIM checks whether the reported mean is possible for any integer dataset before SPRITE asks what the distribution looked like. Always run GRIM first. A GRIM failure is sufficient to flag the statistic; SPRITE adds value when the mean passes GRIM but the M-SD combination still seems anomalous.
GRIMMER	Extends GRIM to standard deviations: checks whether the reported SD is achievable given the reported M, N, and integer constraints, using a more complex calculation. GRIMMER and SPRITE address the same problem from different angles — GRIMMER asks whether SD is possible; SPRITE asks what distribution could produce the M-SD pair together. GRIMMER is faster; SPRITE is more informative.
statcheck	Automated extraction and recalculation of p-values. Operates on test statistics rather than means and SDs; catches a different class of reporting error. Complementary to SPRITE in that both can be applied post-publication using only the published paper.
Multiverse Analysis	If raw data are available, multiverse analysis reveals whether results are robust across defensible analytic choices. SPRITE and GRIM are the post-publication counterparts for integer-scale data: they apply when raw data are unavailable and only summary statistics are reported.

Key Citations

- Heathers, J. A. J., Anaya, J., van der Zee, T., & Brown, N. J. L. (2018). Recovering data from summary statistics: Sample parameter reconstruction via iterative techniques (SPRITE). *PeerJ Preprints*, 6, e26968v1. <https://doi.org/10.7287/peerj.preprints.26968v1>
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STATCHECK

Tier 1 :: Research Methods Errors

At a glance: statcheck automatically extracts reported test statistics from the text of psychology and related research papers — t-tests, F-tests, chi-squared tests, z-tests, and correlations — and recalculates their associated p-values from scratch. It then compares the recalculated p-value to the one reported in the paper, flagging inconsistencies. Errors that cross the $p < .05$ significance threshold are flagged as decision errors — cases where the reported conclusion may be incorrect. No raw data is needed; statcheck works from a machine-readable PDF or text file.

1. Overview

When researchers report the results of statistical tests in the text of a paper, they typically report three numbers: the test statistic, the degrees of freedom, and the p-value. These three numbers are mathematically linked — given any two, the third can be calculated. This means every reported p-value can be independently verified from the test statistic and degrees of freedom alone, without access to the original data.

statcheck, developed by Michèle Nuijten and colleagues at Tilburg University and first released in 2014, automates this verification at scale. It scans a paper's text using pattern-matching rules that recognize standard reporting formats (e.g., $t(38) = 2.31, p = .026$), extracts the test type, statistic, and degrees of freedom, recalculates the p-value using the appropriate distribution function, and checks whether the reported p-value is consistent with the recalculated one.

The tool was designed for the specific reporting conventions of the American Psychological Association (APA) style, which require test statistics to be reported in a standardized format. This constraint is simultaneously its greatest strength — it enables fully automated checking — and its key limitation: it only catches errors in statistics that follow the recognized format, and misses errors in non-standard reporting or in statistics that are not in the extracted set.

statcheck does not detect fabrication or intentional manipulation. A researcher who invents a test statistic and a consistent p-value will pass statcheck without error. What it detects is inconsistency — cases where the reported p-value does not match the test statistic and degrees of freedom, whether due to rounding, transcription error, copy-paste mistakes, or the more concerning possibility of post-hoc adjustment of reported values.

2. Methodological Overview

Extraction via regular expressions

statcheck uses a set of regular expressions — pattern-matching rules applied to paper text — to identify and extract reported test results. The recognized formats cover the most common APA-style test statistics:

t-test: $t(df) = \text{value}, p = \text{value}$

Example: $t(48) = 2.14, p = .037 \rightarrow$ statcheck extracts: type=t, df=48, stat=2.14, p_reported=.037

F-test: $F(df1, df2) = \text{value}, p = \text{value}$

Example: $F(2, 87) = 4.32, p = .016 \rightarrow$ extracts: type=F, df1=2, df2=87, stat=4.32, p_reported=.016

Chi-squared: $\chi^2(df) = \text{value}, p = \text{value}$

Also accepts $r(df) = \text{value}, p = \text{value}$ for correlations and $z = \text{value}, p = \text{value}$ for z-tests.

Recalculation and comparison

For each extracted result, statcheck recalculates the p-value using the standard distribution function in R for that test type: `pt()` for t-statistics, `pf()` for F-statistics, `pchisq()` for chi-squared, `pnorm()` for z-scores. The recalculated p-value is then compared to the reported p-value.

Two thresholds are applied:

Inconsistency	The recalculated p-value differs from the reported p-value by more than would be expected from rounding alone (default threshold: the reported p-value is
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	inconsistent if it deviates from the recalculated value at the second decimal place or beyond). Roughly half of detected inconsistencies are benign rounding errors.
Decision error	The recalculated p-value crosses the $p = .05$ significance threshold in the opposite direction from the reported p-value. For example: reported $p = .049$ (significant) but recalculated $p = .051$ (non-significant), or vice versa. Decision errors are the subset of inconsistencies that would change the reported statistical conclusion. These are the primary focus of integrity analysis.

The two-tier classification is important: most inconsistencies are not decision errors. A reported $p = .023$ recalculating to $p = .024$ is an inconsistency but not a decision error. Only inconsistencies where the significance threshold is crossed in the wrong direction warrant concern about whether the reported conclusion is correct.

3. Example Application: Psychology Literature Survey

Nuijten et al. (2016) applied statcheck to a corpus of 30,717 articles published in eight major psychology journals between 1985 and 2013. The analysis extracted and checked 258,050 null hypothesis significance tests. The results were striking:

- Approximately 49% of papers contained at least one reporting inconsistency — a recalculated p-value that did not match the reported one.
- Approximately 13% of papers contained at least one decision error — an inconsistency where the significance conclusion would change.
- Decision errors were not random: they were significantly more likely to favor the researcher's hypothesis (a non-significant result reported as significant) than to work against it. This asymmetry is consistent with selective reporting or post-hoc adjustment, not random transcription error.
- Error rates decreased slightly in recent years but remained substantial throughout the study period. Journals with more rigorous statistical reporting policies had modestly lower error rates.

The paper triggered significant discussion about the reliability of psychological research reporting and prompted several journals to implement statcheck screening at the submission stage. Some journals now run statcheck automatically on all submitted manuscripts and require authors to explain any flagged inconsistencies before peer review.

A follow-up study by Epskamp and Nuijten (2016) applied statcheck to the Open Science Collaboration's Reproducibility Project dataset, finding that decision errors were significantly more common in studies that failed to replicate than in those that successfully replicated — consistent with the hypothesis that some decision errors reflect meaningful manipulation of reported statistics rather than innocent rounding.

4. Requirements & Applicability

statcheck is notable for its low barrier to entry — it can be applied to any paper that reports statistics in a recognizable format, with no access to raw data required.

Machine-readable text	statcheck requires the paper's text in a format that can be parsed by R. PDF files can be processed directly by statcheck via PDF-to-text conversion (the tool handles this internally). Scanned image PDFs — common in older journal archives — cannot be processed and must be OCR-converted first.
APA-style reporting format	statcheck recognizes statistics reported in APA format: test statistic, degrees of freedom in parentheses, equals sign, value, p-value. Non-standard formats (e.g., 'the result was significant at $p < .05$ ' without reporting the test statistic) are not extracted and not checked. Articles in fields that do not follow APA reporting conventions will have low extraction rates.
Test types supported	t, F, r (correlation), χ^2 (chi-squared), and z. Results from regression coefficients, Bayesian analyses, multilevel models, structural equation models, mixed-effects models, and most non-parametric tests are not extracted unless they are expressed in terms of a supported test statistic.

Raw data required?	No. statcheck operates entirely on the text of the published paper. This is its key practical advantage and the reason it can be applied at scale to large literature corpora.
Language	The pattern-matching rules are designed for English-language papers. Non-English papers or papers with unconventional typographic conventions (e.g., commas as decimal separators in some European journals) may have lower extraction rates or higher false-positive rates.
One-tailed vs. two-tailed	statcheck assumes two-tailed tests by default. One-tailed p-values that are reported as such will appear as inconsistencies because they are half the two-tailed value. The tool provides an option to check one-tailed results explicitly, but this must be applied carefully.

5. Error Taxonomy: What statcheck Finds and What It Misses

Understanding statcheck's output requires distinguishing between the types of errors it can and cannot detect. This matters for interpreting a clean statcheck result — it is not a certification of accuracy, only of consistency in the checked statistics.

What statcheck can detect

- Transcription errors: a test statistic, degrees of freedom, or p-value that was mistyped when the paper was written or typeset.
- Copy-paste errors: statistics copied from statistical software output but with a value from the wrong cell or row.
- Rounding errors: a p-value rounded to fewer decimal places than the test statistic justifies, causing a reported $p = .05$ for a test that actually gives $p = .054$.
- Decision errors: the subset of inconsistencies large enough to change the significance conclusion. These may reflect any of the above causes, or may reflect more deliberate adjustment of a marginal result.

What statcheck cannot detect

- Fabricated but internally consistent statistics: a researcher who invents a t-statistic of 2.31, $df = 38$, and correctly calculates $p = .025$ will produce no inconsistency. statcheck verifies the math, not the underlying data.
- Errors in non-extracted statistics: only test results in recognized formats are checked. A paper may have dozens of unextracted statistics that contain errors.
- Wrong test applied: a researcher who applies a paired t-test to independent samples, or uses the wrong degrees of freedom for an ANOVA, will produce internally consistent statistics that are nonetheless incorrect. statcheck has no way to assess whether the right test was used.
- Effect size errors: Cohen's d, eta-squared, and other effect sizes are not extracted or checked by statcheck.
- Selective reporting: statcheck cannot tell whether the extracted statistics represent all tests conducted or only the subset that was significant. A paper that reports five significant results from fifty conducted analyses will pass statcheck cleanly on all five reported tests.

6. Limitations

- The false positive rate for decision errors is non-trivial. Nuijten et al. estimated that approximately 1.6% of all extracted results are decision errors — but in any individual paper with a small number of results, even a single flagged decision error should be interpreted cautiously. Rounding and transcription error are far more common than deliberate manipulation.
- Extraction rate varies substantially across papers. Papers with many statistics reported in tables (rather than text) or in non-standard formats will have low extraction rates. A paper with zero extracted results cannot be evaluated; this is not evidence of accuracy.
- statcheck does not evaluate the appropriateness of the statistical test used, the validity of the degrees of freedom, or whether the reported test matches the one described in the methods section. These require human review.

- The tool's focus on p-values is a limitation in an era of increasing effect size and confidence interval reporting. As journals shift to requiring confidence intervals and discouraging binary p-value reporting, statcheck's coverage of reported results will decrease.
- Checking at scale requires careful interpretation of aggregate results. A 13% decision error rate across a literature does not mean 13% of papers contain deliberate falsification — the vast majority reflect mundane errors. Individual paper-level results should never be publicly attributed to misconduct without further investigation.

7. How to Apply

Web interface (statcheck.io)

The quickest way to check a single paper is to paste its text into the statcheck web app at statcheck.io. The app returns a table of extracted statistics with recalculated p-values, inconsistency flags, and decision error flags. No installation required.

R package

- Install: `install.packages('statcheck')` in R.
- Check a single PDF: `result <- statcheck('path/to/paper.pdf')`. Returns a data frame with one row per extracted statistic.
- Check a folder of PDFs: `result <- statcheckPDFFolder('path/to/folder/')`. Scales to hundreds of papers automatically.
- Check pasted text: `result <- statcheck('t(38) = 2.14, p = .037', text = TRUE)`. Useful for testing specific results without processing a full PDF.
- Inspect results: result shows columns for Source, Statistic, df1, df2, Test value, p_reported, p_computed, Reported_conclusion, Correct, Error, and Decision_error. Filter on `Decision_error == TRUE` for the most consequential flags.

Interpreting output

A clean statcheck output (no inconsistencies) is not a certification of accuracy — it means all extracted statistics are internally consistent. A flagged decision error warrants re-examination of the original analysis and, where possible, re-running the test with the reported data. Authors should be contacted for clarification before any public allegation of error is made.

8. Related Tools

GRIM Test	Checks whether reported means are arithmetically possible given integer response data and sample size. Complementary to statcheck: GRIM targets means while statcheck targets test statistics. Both operate on published summary statistics without raw data.
SPRITE	Reconstructs possible underlying datasets from reported M, SD, and N. Deeper than statcheck but requires integer-scale data. Together, GRIM, SPRITE, and statcheck form a comprehensive Tier 1 screening toolkit for published results.
p-curve	Uses the distribution of significant p-values to assess evidential value across a literature. Where statcheck flags individual inconsistent p-values within papers, p-curve assesses whether the pattern of significant p-values across papers reflects genuine effects or selective reporting.
Registered Reports	The structural prevention: if study outcomes are committed to in advance, the post-hoc adjustment of p-values that statcheck detects becomes impossible. Journals requiring Registered Reports show substantially lower decision error rates in statcheck analyses.

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THE P-CURVE

Diagnosing Evidential
Value in a Literature

At a glance: The p-curve examines the distribution of statistically significant p-values (those between 0 and .05) across a set of studies. Genuine effects produce a right-skewed distribution — many p-values clustered near zero. A flat or left-skewed curve is a diagnostic signature of p-hacking or an absent true effect. The method requires only the test statistics from a set of significant results — no raw data needed.

1. Overview

The p-curve, introduced by Uri Simonsohn, Leif Nelson, and Joseph Simmons in 2014, is a tool for diagnosing the evidential value of a set of published findings. Its central insight is deceptively simple: if a true effect exists and studies are conducted with adequate power, most significant p-values will cluster near the low end of the 0–.05 range — not spread evenly across it, and not piled up just below .05.

When researchers p-hack — running many analyses and reporting only the one that crossed the threshold — they tend to produce p-values that are just barely significant: concentrated near .04 or .049 rather than near .001 or .01. This selective process leaves a distinctive fingerprint in the distribution of published p-values. The p-curve reads that fingerprint.

The method addresses a practical gap: where multiverse analysis requires access to raw data to diagnose fragility prospectively, p-curve can be applied retrospectively to an entire literature using only the test statistics reported in published papers. This makes it one of the most accessible meta-scientific tools for graduate students and researchers without data access.

2. Methodological Overview

The analyst collects the focal hypothesis test from each study in a set — the single test most directly tied to the paper's primary claim. These test statistics (t , F , z , χ^2 , r) are converted to exact p-values and plotted as a distribution across the 0–.05 range.

Two diagnostic tests are then applied:

Right-skew test: H_0 = uniform distribution across 0–.05

If the distribution is right-skewed (many p-values near 0), the test rejects H_0 — indicating evidential value. The test uses a half-normal approximation.

Flatness test: H_0 = 33% power distribution (pp-values)

Tests whether the p-curve is flatter than expected under studies with 33% power. A flat result relative to this benchmark indicates p-hacking without a true effect.

The 'pp-value' transformation is central to the method: each p-value is transformed into the probability of observing a p-value at least as small, given a specific alternative hypothesis. Under the null of no true effect combined with p-hacking, this transformed distribution is uniform; under a genuine effect, it skews right.

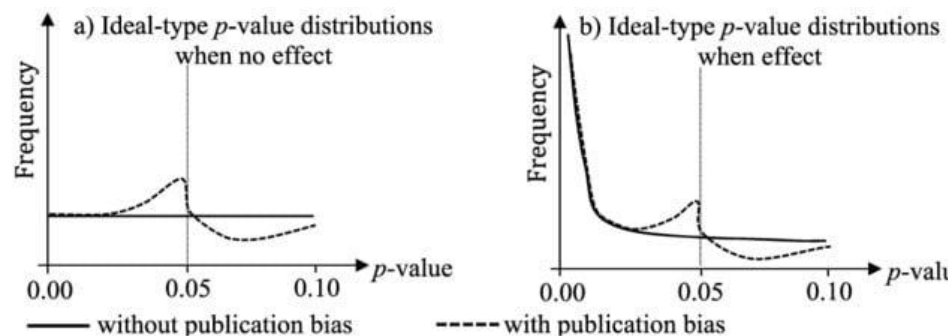
Simonsohn et al. recommend testing for both right-skew and flatness — a curve that is right-skewed passes the first test (evidential value exists) and must also pass the second test for the evidential value to be considered robust. A flat curve that fails the second test is consistent with widespread p-hacking that incidentally produces just-significant results.

The p-curve app (pcurve.com) automates all calculations. The analyst enters test statistics directly and the app produces the curve, the two tests, and an estimate of the average statistical power underlying the studies — a useful additional output when comparing literatures.

3. Interpreting the Curve Shape

Reading a p-curve is partly visual and partly inferential. Three canonical shapes matter:

Right-skewed	Many p-values near 0, few near .05. Signature of genuine effects with adequate power. The literature has evidential value — effects are real, though they may be smaller than the published estimates suggest if publication bias is also present.
Flat / uniform	p-values spread roughly evenly across 0–.05. Consistent with p-hacking in the absence of a true effect: researchers ran many tests and reported whichever crossed the line, producing a uniform distribution rather than the right-skew a real effect would generate.
Left-skewed	p-values piled up near .049. The strongest signature of p-hacking: studies are barely significant because researchers stopped as soon as they crossed the threshold. Reported effects almost certainly overstate or fabricate the underlying relationship. Also called a 'left-skewed p-curve'.



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4. Example Application: Ego Depletion

The ego depletion literature — the highly influential finding that acts of self-control deplete a limited mental resource — was a prominent target of p-curve analysis. Simonsohn and colleagues applied p-curve to the set of studies Hagger et al. (2010) used in a widely-cited meta-analysis supporting the effect.

The resulting p-curve was notably flat and partially left-skewed, failing the evidential value test. The analysis indicated that the published significant findings were consistent with p-hacking rather than a genuine underlying effect. This was a striking result because the individual studies had each passed peer review and the meta-analytic effect size appeared substantial.

The p-curve finding preceded by two years the large pre-registered multi-lab replication by Hagger et al. (2016) — across 23 labs — which found $d \approx 0.04$, essentially zero. The p-curve had correctly diagnosed the absence of evidential value from the published record alone, without requiring any new data collection.

5. Requirements & Applicability

The p-curve has specific requirements that govern when and how it can be applied appropriately.

Test statistics	The exact test statistic (t , F , z , χ^2 , r) and degrees of freedom for each study's focal test. These must be recoverable from the paper — p-values alone are insufficient because they cannot be back-transformed to exact statistics without df.
Focal test selection	One test per study — the test most directly linked to the primary theoretical claim. Multi-study papers should contribute one test per study, not one per

	paper. This is the most consequential analyst decision and must be documented.
Significant results only	The p-curve is computed using only significant results ($p < .05$). Non-significant results are excluded from the curve itself, though their presence is noted as context. This is by design — the method diagnoses the shape of the significant portion.
Minimum study count	Reliable inference requires approximately 20–25 studies. With fewer than 10 significant results the curve is too noisy for the formal tests, though visual inspection may still be informative.
Raw data required?	No. The method operates entirely on reported test statistics in published papers. This is p-curve's key practical advantage over multiverse and specification curve analysis.
Consistent hypothesis	All studies should be testing the same directional hypothesis. Mixing studies that test different effects or directions produces an uninterpretable aggregate curve.
No pooling of effects	p-curve is not a substitute for meta-analysis — it does not estimate the pooled effect size. It diagnoses evidential value, not magnitude. The p-curve app does report a power estimate, which implies an effect size, but this is secondary and should be interpreted cautiously.

6. Limitations

- p-curve diagnoses the significant portion of the literature only. It cannot speak to unpublished null results or the severity of publication bias — a right-skewed curve is consistent with genuine effects even in a severely publication-biased literature.
- The focal test selection is consequential and subjective. Different analysts may disagree on which test represents the primary claim of a paper, producing different curves from the same literature. Sensitivity analyses across reasonable alternative focal tests are recommended.
- The method assumes all studies test the same directional hypothesis with reasonable power. Mixing underpowered studies, different operationalizations, or different populations can distort the curve shape independently of p-hacking.
- p-curve cannot distinguish p-hacking from a true effect that is very small and the studies are severely underpowered — both produce flat curves. The flatness test tests against a 33% power benchmark, which may be too high or too low for some literatures.
- The power estimate derived from the p-curve rests on several assumptions and should not be treated as a precise estimate. It provides a rough diagnostic, not a substitute for a formal power analysis.
- p-curve applies only within the 0–.05 significance range. Results reported as exact p-values below .001 are often truncated in papers (reported as ' $p < .001$ ') and must be entered conservatively, which can affect the curve shape.

7. How to Apply

Using the p-curve app (pcurve.com)

The web app maintained by Simonsohn, Nelson, and Simmons is the standard implementation. It accepts test statistics entered in a simple text format (e.g., ' $t(38) = 2.31$ ' or ' $F(1,76) = 5.41$ ') and produces:

- The p-curve plot with the two diagnostic test results
- A statistical power estimate for the studies in the set
- Downloadable results for reporting

Step-by-step procedure

- Identify the set of studies: define inclusion criteria before examining the results — typically all direct tests of the focal hypothesis in the literature.

- For each study, identify the single focal test — the statistical test most directly tied to the primary theoretical claim. Document the selection rule explicitly.
- Collect the exact test statistic and degrees of freedom from each paper. If a paper reports only $p < .001$, enter as $p = .001$ (conservative). Exclude non-significant results.
- Enter each statistic into the p-curve app and run the two tests. Report both the right-skew test and the flatness test, along with the visual curve.
- Conduct sensitivity analyses: re-run the curve with plausible alternative focal test selections to check whether conclusions are robust to this choice.

R implementation

The dmetar R package (Harrer et al. 2021) includes a `pcurve()` function that accepts a vector of p-values and produces the curve and tests. This is useful for batch processing of large literature sets and for integration with meta-analytic workflows.

8. Related Tools

p-uniform	A complementary p-value distribution method that also operates on significant p-values but models the full distribution under a specific assumed effect size. More sensitive to heterogeneity than p-curve; the two methods should converge when both assumptions are met.
Caliper test	Tests for an excess of results just below the $p = .05$ threshold relative to just above it — a direct signature of stopping-rule manipulation or rounding. Less powerful than p-curve but simpler and requires only a count of results above and below the threshold.
z-curve 2.0	Extends the p-value distribution approach to include non-significant results, estimating the expected replication rate and discovery rate across the full distribution of test statistics. Complements p-curve by working with the complete set rather than only the significant portion.
Excess significance test (TES)	Compares the observed count of significant results to the count expected given study power. Where p-curve diagnoses the shape of the significant distribution, TES diagnoses whether there are too many significant results in the first place.
Funnel plot / Egger test	Meta-analytic tools that detect publication bias by examining asymmetry in effect size vs. precision plots. Complementary to p-curve: funnel asymmetry reflects missing studies; a left-skewed p-curve reflects within-study manipulation.

Key Citations

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P-UNIFORM

Tier 2a :: p-Hacking / Selective Reporting

At a glance: p-uniform is a meta-analytic method that estimates the true underlying effect size in a literature and tests for publication bias by modelling the full conditional distribution of significant p-values under a specific assumed effect. Unlike p-curve, it produces a bias-corrected effect size estimate alongside an evidential value test. It requires only the test statistics from significant results — no raw data needed.

1. Overview

p-uniform was developed by Robbie van Aert, Marcel van Assen, and Jelte Wicherts, with initial versions published around 2015. It belongs to the same family as p-curve — both methods work exclusively with the significant portion of a literature's p-values — but it takes a fundamentally different approach to what it tries to estimate.

Where p-curve asks 'does this set of results contain genuine evidential value, and if so, how much statistical power do the studies have?', p-uniform asks 'what true effect size would have generated exactly this distribution of significant p-values?' It inverts the selection model: given that we know only significant results were published, and given that we can model the probability of achieving significance as a function of the true effect, we can back-calculate the effect size that makes the observed p-value distribution most probable.

This makes p-uniform particularly valuable in two situations: when a researcher wants a bias-corrected effect size estimate from a biased literature, and when the heterogeneity assumptions required by other bias-correction methods (trim-and-fill, PET-PEESE) are poorly met. p-uniform* (the asterisk version, van Aert & van Assen 2021) extends the original method to handle heterogeneity directly, making it one of the most statistically sophisticated tools in the publication bias toolkit.

2. Methodological Overview

The core insight of p-uniform is that if a true effect δ exists and studies are conducted with varying sample sizes, the distribution of significant p-values is not uniform — it is concentrated toward smaller p-values for studies with higher power, and toward larger p-values (just below .05) for studies with lower power. Knowing this expected distribution, and comparing it to what is actually observed, allows both a test of whether $\delta = 0$ and an estimate of the most likely non-zero δ .

The uniform quantile transformation

For each study i with sample size n_i and observed test statistic t_i , p-uniform computes the conditional p-value: the probability of observing a test statistic at least as extreme as t_i , given that the study was significant ($p < .05$) and given a specific true effect δ . Under the correct effect size, these conditional p-values should be uniformly distributed between 0 and 1.

$$q_i(\delta) = P(T \geq t_i \mid p < .05, \delta) / P(p < .05 \mid \delta)$$

The conditional p-value for study i . Under the true effect δ , $q_i(\delta) \sim \text{Uniform}(0,1)$. Deviation from uniformity indicates either a wrong assumed effect or publication bias.

$$H_0 \text{ test: } q_i(\delta=0) \sim \text{Uniform}(0,1)$$

The publication bias / evidential value test: if results are consistent with $\delta = 0$ (no true effect), the conditional p-values under $\delta = 0$ should be uniform. Right-skew rejects H_0 , indicating evidential value.

$$\text{Estimate } \hat{\delta}: \text{ find } \delta \text{ such that } \text{median}(q_i(\delta)) = 0.5$$

The effect size estimate: the value of δ that makes the observed conditional p-value distribution most uniform, i.e., closest to $\text{Uniform}(0,1)$. This is the method-of-moments estimator.

The test statistic is computed using a sum of log-transformed conditional p-values (analogous to Fisher's method), which follows a chi-squared distribution under the null. The estimator is solved numerically — the R package puniform implements this efficiently for t , F , and r statistics.

p-uniform* — the heterogeneity extension

The original p-uniform assumes a fixed true effect (all studies estimate the same δ). When true effects are heterogeneous across studies — different populations, interventions, or contexts — the conditional p-value distribution is no longer uniform even under the correct average effect, because variance in δ creates variance in the conditional distribution.

p-uniform* (van Aert & van Assen 2021) incorporates between-study variance τ^2 directly into the selection model, estimating both the mean effect μ and the variance τ^2 simultaneously. This makes it applicable to heterogeneous literatures where the fixed-effect assumption of original p-uniform would be violated, at the cost of requiring more studies for stable estimation. Under homogeneity, p-uniform* collapses to p-uniform.

3. Example Application: Ego Depletion Revisited

Carter et al. (2015) applied both PET-PEESE and p-curve to the ego depletion literature — the widely cited finding that exerting self-control depletes a limited cognitive resource. p-uniform, applied to the same set of studies, produced a bias-corrected effect size estimate of approximately $d = 0.21$, compared to the raw meta-analytic estimate of $d = 0.62$.

What makes this example instructive is the comparison across methods: PET-PEESE estimated $d \approx 0.24$, p-curve estimated average power of approximately 31% (implying $d \approx 0.20$ – 0.25), and p-uniform landed at $d \approx 0.21$. The convergence of three different bias-correction methods on a similar, much lower estimate than the published meta-analysis is strong evidence that the literature is substantially inflated — and that these tools, despite their different mathematical foundations, are picking up the same underlying signal.

The subsequent 23-lab pre-registered replication (Hagger et al. 2016) found $d \approx 0.04$, essentially indistinguishable from zero — suggesting all three bias-corrected estimates still overcorrected in the positive direction, but were far closer to the truth than the uncorrected meta-analytic estimate of $d = 0.62$.

4. Requirements & Applicability

p-uniform shares most of its data requirements with p-curve but adds some additional constraints related to its parametric assumptions.

Test statistics and df	The exact test statistic (t , F , z , r) and degrees of freedom for each study's focal result. p-values alone are insufficient — they cannot be back-transformed to the conditional distribution without the underlying statistic and df.
Significant results only	Like p-curve, p-uniform is computed using only significant findings ($p < .05$). Non-significant results are excluded from the estimation. This is by design — the method models the selection mechanism that produced only significant results.
One test per study	A single focal test per study, corresponding to the primary theoretical claim. Averaging or aggregating multiple tests within a study is not supported and will distort the conditional distribution.
Consistent effect size metric	All studies must be estimating the same effect in the same direction. p-uniform cannot be applied to a mixed literature where some studies test effect A and others test effect B, even if nominally related. Effect size heterogeneity of the random-effects type is handled by p-uniform* but requires at least 20–30 studies.
Minimum study count	Reliable estimation requires approximately 10–15 studies for p-uniform (fixed-effect). p-uniform* requires approximately 20–30 for stable variance estimation. With fewer studies the confidence interval on δ is very wide.
Raw data required?	No. The method operates entirely on published test statistics. This is the key practical advantage shared with p-curve over multiverse and specification curve analysis.
Effect size type	The puniform R package implements p-uniform for Cohen's d (from t -tests), partial η^2 (from F -tests), and Pearson's r . Odds ratios and hazard ratios from logistic/survival models are not currently supported natively.

5. p-uniform vs. p-curve: When to Use Which

p-uniform and p-curve are close relatives that answer related but distinct questions. The choice between them depends on what you want to know.

Primary output	p-curve: evidential value test + average power estimate. p-uniform: bias-corrected effect size δ + evidential value test.
Effect size estimate	p-curve produces an effect size estimate only indirectly (via power). p-uniform produces δ directly as its primary output — this is its key advantage when a bias-corrected magnitude estimate is needed.
Heterogeneity handling	p-curve treats heterogeneity informally (it affects curve shape but is not modelled). p-uniform* models τ^2 explicitly — preferred when heterogeneity is known or expected to be substantial.
Sample size sensitivity	p-uniform is sensitive to the accuracy of reported sample sizes. p-curve is less so, because it uses the full p-value distribution rather than a parametric conditional distribution.
Recommended use	Use p-curve when diagnosing evidential value qualitatively (is there signal here?). Use p-uniform when you need a publishable bias-corrected effect size estimate and can verify the fixed-effect assumption is reasonable.
Convergence check	Best practice is to run both. If they converge on similar conclusions, confidence in the result increases substantially. Divergence signals model sensitivity and should be reported.

6. Limitations

- p-uniform assumes publication bias takes a specific form — that all studies with $p < .05$ are published and all with $p \geq .05$ are suppressed. In reality, bias is partial and probabilistic. Violation of this sharp selection assumption biases the δ estimate, typically toward overestimation of the true effect.
- The fixed-effect assumption of original p-uniform (all studies estimate the same δ) is violated in most real literatures. p-uniform* addresses this but requires more studies and has higher computational cost. Using fixed-effect p-uniform in a heterogeneous literature produces biased estimates.
- p-uniform is sensitive to focal test selection. If an analyst selects a non-focal test (e.g., a secondary outcome or a subgroup analysis) instead of the primary test, the conditional distribution is misspecified and the estimate is uninformative.
- The method cannot detect p-hacking that operates within rather than between studies — e.g., a researcher who runs many internal analyses and reports the most significant one. The published p-value is significant, which is all p-uniform sees, but it does not represent the correct comparison to the theoretical distribution.
- p-uniform has lower statistical power than p-curve for the evidential value test when the true effect is large. p-curve's right-skew test is more sensitive to the skewness signal that a large true effect produces; p-uniform's uniformity test is more sensitive to the mean effect.
- The numerical optimizer for δ can be unstable when the number of studies is small or when effect sizes are very heterogeneous. Always inspect the likelihood function and confidence interval, not just the point estimate.

7. How to Apply

R package: puniform

The puniform R package (van Aert 2022) is the standard implementation. It accepts vectors of test statistics and sample sizes and returns the evidential value test, the bias-corrected effect size estimate, its confidence interval, and the p-uniform* estimates if requested.

Minimal workflow

- Identify the literature: define inclusion criteria before examining results. Collect the focal test statistic (t, F, or r) and sample sizes for each significant finding.
- Install the package: `install.packages('puniform')` in R. Load with `library(puniform)`.

- Run the basic analysis: `puniform(tobs = c(2.3, 2.8, 3.1, ...), n = c(40, 60, 35, ...), side = 'right')` for a set of one-sided t-tests. Adjust the 'method' argument to 'P' (p-uniform) or 'LNP' (log normal p, more robust) as needed.
- For heterogeneous literatures, use the `puni_star()` function from the same package to fit p-uniform*, which estimates both μ and τ^2 .
- Run p-curve alongside using the p-curve app (pcurve.com) and compare conclusions. Report both sets of results and note any divergence.

Reporting standards

Report: the number of studies included, the focal test selection rule, the evidential value test statistic and p-value, the δ estimate with 95% confidence interval, whether fixed-effect or random-effects (p-uniform*) was used and why, and the p-curve result for comparison. If the two methods diverge, discuss which assumptions are more likely violated in this literature.

8. Related Tools

p-curve	The closest relative: same input requirements, same significant-results-only design, same inability to handle non-significant results. p-curve diagnoses evidential value qualitatively and estimates average power; p-uniform estimates the bias-corrected effect size directly. Run both and compare.
PET-PEESE	Meta-regression approach that also produces a bias-corrected effect size estimate, but uses all studies (significant and non-significant) and models the relationship between effect size and standard error. PET-PEESE requires no distributional assumptions about the selection mechanism but is sensitive to heterogeneity in a different way. The two methods should converge in well-behaved literatures.
Trim-and-Fill	Non-parametric imputation approach that also produces a bias-corrected pooled estimate. Less theoretically principled than p-uniform but widely reported and easily understood. Convergence of trim-and-fill, PET-PEESE, and p-uniform on similar estimates substantially strengthens confidence in the bias-corrected value.
z-curve 2.0	Extends the p-value distribution approach to include non-significant results, estimating expected replication and discovery rates. Complements p-uniform by providing a field-level summary that does not require the significant-results-only assumption.
Excess Significance Test (TES)	Tests whether the observed count of significant results exceeds what would be expected given estimated study power. Complements p-uniform's within-distribution test with an across-literature count test.

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THE CALIPER TEST**Tier 2a :: p-Hacking / Selective Reporting**

At a glance: The caliper test detects an excess of statistically significant results that fall just below a reporting threshold — most commonly $p < .05$ — relative to results that fall just above it. A spike near the threshold is a signature of selective stopping, optional stopping rules, or rounding of p-values to achieve significance. The test requires only reported p-values or test statistics from a set of studies; no raw data is needed.

1. Overview

The caliper test was introduced by Alan Gerber and Neil Malhotra in 2008 in the context of empirical sociology, where they observed an implausible discontinuity in the distribution of published z-scores around the conventional significance threshold. The intuition behind the test is a direct extension of the observation that if researchers are free to choose when to stop collecting data, which covariates to include, or how to round reported statistics, they will systematically produce results that land just on the right side of the threshold rather than just above it.

Under honest research practice, the distribution of test statistics near any threshold should be smooth. If a true effect exists, significant and near-significant results should differ only by sampling variation, and results just below .05 should be approximately as common as results just above it. A sharp spike below the threshold — and a corresponding trough just above it — is consistent with only two explanations: strategic manipulation of the analysis process, or systematic rounding of p-values that are close to but above .05 down to just below it.

The caliper test is deliberately simple and conservative. It does not model the full distribution of p-values (as p-curve and z-curve do), and it does not require assumptions about the true underlying effect size (as p-uniform does). It makes only one assumption: that in the absence of manipulation, the count of results just below the threshold should not significantly exceed the count just above it within a narrow symmetric window. This simplicity is both its strength — it is robust to many distributional assumptions — and its limitation, since it discards information from the full distribution.

2. Methodological Overview

The analyst defines a threshold (typically $p = .05$) and a caliper — a symmetric window of fixed width around that threshold. The caliper captures results in two bins: those falling between the threshold and the lower boundary of the caliper window (just significant), and those falling between the threshold and the upper boundary (just non-significant). The test asks whether the count in the just-significant bin significantly exceeds the count in the just-non-significant bin.

Basic binomial test

In the simplest implementation, the counts in the two bins are compared using a one-tailed binomial test. Under the null hypothesis of no manipulation, results are equally likely to fall in either bin, so the expected split is 50/50. An excess of results just below the threshold produces a binomial p-value indicating the probability of observing at least that many results in the lower bin by chance.

`n_below = count of results in (threshold - caliper, threshold]`

e.g., for threshold = .05 and caliper = .01: results with p in (.04, .05]

`n_above = count of results in [threshold, threshold + caliper)`

e.g., for threshold = .05 and caliper = .01: results with p in [.05, .06)

`H0: P(result falls below threshold) = 0.5`

Test statistic: $z = (n_below - n_above) / \sqrt{n_below + n_above}$. Under H_0 , $z \sim N(0,1)$. A significantly positive z indicates excess just-significant results.

Caliper width selection

The caliper width is the most consequential analyst decision. A very narrow caliper (e.g., .005) captures only the most extreme rounding and is highly sensitive to the specific manipulation being detected but has low statistical power with small samples. A wide caliper (e.g., .025) has more power but captures results that

may have arisen for legitimate reasons other than manipulation — a study with a true effect near the threshold will naturally produce results in the window with some frequency.

Gerber and Malhotra (2008) recommended a caliper of .025 (capturing results in the range .025 to .050 and .050 to .075) for their initial application. Subsequent researchers have recommended conducting sensitivity analyses across multiple caliper widths and checking whether the excess is consistent across them — a spike that appears only at one specific caliper width is more likely to be a statistical artifact than a genuine manipulation signal.

Regression discontinuity extension

A more powerful variant, developed by Egger and colleagues and applied extensively in economics, treats the p-value distribution as a continuous function and tests for a discontinuity at the threshold using local polynomial regression. This approach uses more of the distribution's information than the simple bin comparison and allows formal estimation of the size of the discontinuity — the excess density at the threshold — rather than just its statistical significance.

$$f(p) = \alpha + \beta_1 p + \beta_2 p^2 + \gamma \cdot N(p \leq \text{threshold}) + \epsilon$$

Local polynomial regression on the p-value density. The discontinuity estimate γ measures the excess density at the threshold. A significantly positive γ indicates a spike consistent with selective reporting.

3. Example Application: Sociology and Political Science Journals

Gerber and Malhotra's original 2008 paper applied the caliper test to articles published in the American Journal of Sociology and the American Journal of Political Science between 1995 and 2005. They converted all reported z-scores, t-statistics, and F-statistics to equivalent two-tailed p-values and binned the results within a caliper of .025 around $p = .05$.

The results were striking: there were approximately 2.5 times as many results in the just-significant bin (.025 to .050) as in the just-non-significant bin (.050 to .075). Across both journals combined, the binomial test was highly significant. The distribution showed a pronounced spike immediately below the threshold, with a corresponding trough immediately above it — consistent with widespread optional stopping or threshold-seeking behavior in both disciplines.

A second analysis in the same paper examined the $p = .10$ threshold, which some authors treat as marginally significant. A smaller but still substantial spike appeared there as well — consistent with researchers targeting multiple thresholds depending on the strength of their effect and the conventions of their subfield.

Subsequent applications by Head et al. (2015, PLOS Biology) found similar spikes across a sample of over 100,000 p-values extracted from the biomedical and behavioral literature using automated text mining. The pattern was not uniform across fields: experimental fields with pre-registered hypotheses showed smaller spikes than observational fields with more flexible analytic choices. Caliper analysis has since been applied to economics, education research, management, and clinical trial reporting, consistently finding evidence of threshold-targeting behavior, though the magnitude varies substantially by field and publication venue.

4. Requirements & Applicability

The caliper test's principal advantage is its minimal data requirements. It operates on the marginal distribution of reported p-values, which are available from any study that reports hypothesis tests.

Reported p-values or test statistics	The test requires the exact reported p-value for each hypothesis test, or a test statistic (z, t, F, chi-squared) and degrees of freedom from which an exact p-value can be recovered. Approximate p-values ($p < .05$, $p = \text{NS}$) cannot be used — they provide no information about whether the result fell just above or just below the threshold.
Minimum study count	The binomial test is unreliable with fewer than 20–30 results in the combined window. A caliper of .025 on each side captures a narrow slice of the distribution; many studies produce no results in this window. Practical application typically requires at least 50–100 papers to generate enough observations in the caliper window for meaningful inference.

Consistent threshold	The test is defined relative to a specific threshold. Most applications use $p = .05$, but the test can be applied to any threshold where selective reporting is plausible ($p = .10$, $p = .01$). Analyzing multiple thresholds simultaneously requires multiple comparison correction.
One test per study or all tests?	A consequential analyst choice: using only the focal hypothesis test per study reduces false positives but reduces power; using all reported tests inflates the sample but may mix primary and exploratory results. The choice should be documented and sensitivity analyses conducted across both approaches.
Raw data required?	No. The test operates on reported p-values in published papers. Automated extraction tools (statcheck, custom parsers) can extract p-values from PDF text at scale, making the caliper test one of the more computationally scalable detection methods.
Directional hypotheses	Results reported as one-tailed tests should be treated carefully. A one-tailed p of .049 is equivalent in magnitude to a two-tailed p of .098 — combining one- and two-tailed tests without conversion will distort the distribution near the threshold.

5. Limitations

- The caliper test is sensitive to the choice of caliper width. Different widths can produce qualitatively different conclusions from the same dataset. Reporting sensitivity analyses across widths (e.g., .010, .025, .050) is essential but not always done.
- The test cannot distinguish rounding from strategic stopping. A researcher who rounds $p = .052$ to $p = .05$ in a table produces the same distributional signal as a researcher who added data points until p crossed .05. The test detects the signature of threshold-seeking but cannot identify which mechanism produced it.
- Legitimate effects near the threshold produce false signals. If a literature contains many studies of a true effect whose power is near 50%, approximately half of results will fall just below the threshold and half just above — not because of manipulation, but because the effect is real and borderline. Caliper tests applied to such a literature will flag excess just-significant results that are entirely legitimate.
- The test does not account for publication bias in the broader sense. Even without any within-study manipulation, if journals preferentially publish significant results, the caliper window will be depopulated above the threshold (null results withheld) and populated below it (significant results published). The spike signature is produced by selection at the journal level, not necessarily the researcher level. Caliper analysis cannot distinguish these two sources.
- P-value reporting conventions vary across fields and decades. Some journals report p-values to two decimal places (rounding .049 to .05), some to three. This rounding creates artificial spikes at round-number p-values that are not evidence of manipulation. Analyses should check for and account for discretization artifacts.
- The test has low power in small literature samples. With fewer than 50 studies, the caliper window will typically contain too few observations for stable inference. Fields with small literatures — specialized subfields, emerging research areas — are poorly served by this method.

6. How to Apply

Step 1 — Collect and convert p-values

Identify the set of studies (defined by inclusion criteria specified before examining results). For each study, record the focal hypothesis test statistic and convert to an exact two-tailed p-value. If the paper reports only an inequality ($p < .05$), exclude that result from the analysis — it cannot be placed precisely in the distribution. Automated extraction using statcheck or custom PDF parsers can scale this step to hundreds of papers.

Step 2 — Choose threshold and caliper width

Select the threshold of interest (typically $p = .05$) and the caliper width. For an initial analysis, a width of .025 on each side is a reasonable default, capturing results in the ranges (.025, .050] and [.050, .075). Document the choice and plan sensitivity analyses at widths of .010 and .050.

Step 3 — Count results in each bin

Count the number of results falling in the just-significant bin (below the threshold within the caliper) and the just-non-significant bin (above the threshold within the caliper). If analyzing multiple thresholds, repeat for each.

Step 4 — Run the binomial test

Compute the binomial test statistic: $z = (n_{\text{below}} - n_{\text{above}}) / \sqrt{n_{\text{below}} + n_{\text{above}}}$. Compare to the standard normal distribution for a one-tailed test (H_1 : excess below threshold). In R, this is: `binom.test(n_below, n_below + n_above, p = 0.5, alternative = 'greater')`.

Step 5 — Visualize the distribution

Plot a histogram of all p-values in the range [0, 1] with bin widths equal to the caliper (e.g., 0.025). Visual inspection of the spike at the threshold is often more informative than the test statistic alone — it reveals whether the excess is localized at the threshold or part of a broader pattern. The `rddensity` package in R implements the regression discontinuity variant with confidence intervals and plotting functions.

Step 6 — Conduct sensitivity analyses

Re-run the analysis at alternative caliper widths (.010, .050). Test the $p = .10$ threshold to check for secondary threshold-seeking. Compare results using all reported tests versus focal tests only. If conclusions are consistent across these variations, confidence in the finding increases substantially.

7. Caliper Test vs. Related Distribution-Based Tools

The caliper test occupies a specific niche among p-value distribution methods. Understanding where it sits relative to p-curve and z-curve helps clarify when to prefer it.

Caliper test	Tests for a spike at a specific threshold using only results in a narrow window. No distributional assumptions beyond smoothness of the underlying density. Very simple to compute and explain. Low power but highly interpretable. Best for documenting threshold-seeking behavior as a descriptive fact about a literature.
p-curve	Uses the full distribution of significant p-values (0 to .05) to diagnose evidential value. More powerful than the caliper test for detecting the presence or absence of a true effect. Cannot localize the manipulation to the threshold region — a left-skewed p-curve could reflect either p-hacking or a very small true effect.
z-curve 2.0	Uses both significant and non-significant test statistics to estimate replication and discovery rates. Provides field-level summaries not available from the caliper test. More complex, requires more studies, but provides richer output. Complementary: use z-curve for field-level diagnosis, caliper test for threshold-specific documentation.
Egger / funnel plot	Detects publication bias across a meta-analytic set via asymmetry in effect size vs. precision. Operates on effect sizes rather than p-values, making it complementary to caliper analysis. A literature can show caliper-test excess (within-study manipulation) without funnel asymmetry (between-study publication bias), or vice versa.

8. Related Tools

p-curve	Examines the shape of the full significant p-value distribution. Where the caliper test asks 'are there too many results just below the threshold?', p-curve asks 'is the shape of the significant distribution consistent with a true effect?' Run both: caliper test for threshold-specific diagnosis, p-curve for evidential value.
statcheck	Automated extraction and recalculation of p-values from paper PDFs. The primary tool for building the p-value dataset required by the caliper test at scale. <code>statcheck</code>

	also flags decision-changing reporting errors that are distinct from (but correlated with) the threshold-seeking behavior the caliper test detects.
z-curve 2.0	Fits a mixture model to all reported z-scores to estimate expected replication and discovery rates. Uses the full distribution rather than just the threshold window, providing richer output. The two tools are best understood as complementary rather than competing.
Excess Significance Test (TES)	Tests whether the observed count of significant results exceeds what would be expected given study power. Where the caliper test examines the distribution near a threshold, TES examines the overall rate of significance across the literature. Convergence of a positive caliper test and a positive TES provides substantially stronger evidence of systematic reporting problems.
Registered Reports	The prospective remedy: publishing a commitment to report all results regardless of significance removes threshold-seeking incentives at the source. Scheel et al. (2021) showed that Registered Reports produce far fewer results just below $p = .05$ than standard publications — the caliper-test distribution is flatter, as expected from genuine rather than threshold-seeking research.

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Z-CURVE 2.0**Tier 2a :: p-Hacking / Selective Reporting**

At a glance: z-curve 2.0 converts all reported test statistics in a literature to z-scores, then fits a finite mixture model to the full distribution — both significant and non-significant results. From this fit it estimates two distinct quantities unavailable from any other method: the expected replication rate (ERR), which predicts how often significant findings would replicate in direct replications, and the expected discovery rate (EDR), which estimates the proportion of tested hypotheses that are actually true. Requires test statistics and degrees of freedom; no raw data needed.

1. Overview

z-curve 2.0, developed by Frantisek Bartos and Ulrich Schimmack and published in 2022, addresses a fundamental limitation shared by p-curve, p-uniform, and the caliper test: all three methods work exclusively with the significant portion of a literature's p-values, discarding the information contained in non-significant results. This is a substantial loss. Non-significant results, even when unpublished, leave traces — in registered reports, in preregistered studies that report all outcomes, in clinical trial registries, in meta-analytic databases that include non-significant effects. And when they are available, they are highly informative about the statistical properties of the research program that generated the published significant results.

z-curve takes a different approach. Rather than conditioning on significance, it models the full distribution of test statistics — significant and non-significant — as a mixture of noncentral distributions. Each component of the mixture corresponds to a different level of statistical power. By fitting this mixture model to observed z-scores, z-curve simultaneously estimates what the power distribution of the studies looks like and what that implies about the replication prospects of the significant findings and the true discovery rate of the research program.

The two key outputs — ERR and EDR — answer questions that are at the center of the reproducibility debate but that no prior tool could address quantitatively. The ERR answers: if we ran direct replications of all the significant findings in this literature, what fraction would replicate at $p < .05$? The EDR answers: of all the hypotheses being tested in this research program (including the ones that produced non-significant results that were never published), what fraction are actually true? The gap between ERR and EDR — and the gap between EDR and the observed discovery rate in published papers — quantifies the severity of publication bias in the literature.

2. Methodological Overview**Step 1 — Convert all test statistics to z-scores**

Every test statistic (t, F, chi-squared, r) reported in the studies under analysis is converted to an equivalent two-tailed z-score using the inverse normal transformation. For a t-statistic with df degrees of freedom, the conversion is: $z = \text{qnorm}(1 - \text{pt}(|t|, \text{df})/2)$, where pt is the t-distribution CDF. This places all results on a common scale and allows the mixture model to be fit in z-space regardless of the original test type.

The resulting set of z-scores spans from 0 to some upper bound (typically capped at $z = 6$ to exclude extreme outliers that could destabilize the fit). The critical region for the standard alpha = .05 threshold corresponds to $|z| > 1.96$. Results above this threshold are significant; those below are non-significant.

Step 2 — Fit the finite mixture model

The observed z-score distribution is modeled as a mixture of K noncentral half-normal distributions, each representing a cohort of studies conducted with a specific level of statistical power. The mixture has the form:

$$f(z) = \sum^k \pi^k \cdot g(z \mid \lambda^k)$$

Where K is the number of mixture components, π^k are the mixing weights (summing to 1), and $g(z \mid \lambda^k)$ is a noncentral half-normal density with noncentrality parameter λ^k . Each λ^k corresponds to a power level.

$$\text{Power}(\lambda^k) = P(|Z| > 1.96 \mid \lambda^k) = 1 - \Phi(1.96 - \lambda^k) + \Phi(-1.96 - \lambda^k)$$

The power associated with each mixture component. Components with larger λ^k represent high-powered studies; smaller λ^k represent low-powered studies.

The mixture model is fit using maximum likelihood estimation (or EM algorithm) over the observed z-score distribution. The fitting accounts for selection: because non-significant results are less likely to be published, the likelihood is adjusted by the probability of selection under the estimated power distribution. This selection correction is what separates z-curve from naive mixture model fits and what allows it to estimate the full distribution of true power — not just the power of what was published.

Step 3 — Compute ERR and EDR

Once the mixture model is fit and the power distribution estimated, the two key quantities follow directly:

ERR = $\sum \pi^k \cdot \text{Power}(\lambda^k)$

Expected Replication Rate: the weighted average power across all mixture components. Answers: if a direct replication were run for each significant finding, what fraction would replicate at $p < .05$?

EDR = **ERR** / **ODR**

Expected Discovery Rate: ERR divided by the Observed Discovery Rate (the fraction of all tested hypotheses that produced significant results). Answers: what proportion of all tested hypotheses are actually true, including those with non-significant results that were never published?

Bias = **ODR** - **EDR**

Publication bias index: the gap between what was observed in the published record (ODR) and the estimated true discovery rate (EDR). A large positive value indicates severe publication bias.

The z-curve plot — a histogram of all z-scores with the fitted mixture model overlaid — provides a visual summary. The fitted curve shows the expected distribution under the estimated power mix. Deviations of the observed distribution from the fitted curve, particularly the spike near the significance threshold from p-hacking or the trough just above it from publication bias, are visually apparent.

3. Interpreting ERR and EDR

The relationship between ERR and EDR reveals the statistical health of a research program in a way that no single number can capture. Four canonical patterns are worth understanding:

High ERR, High EDR	A well-powered research program with genuine effects and low publication bias. Studies are adequately powered, most tested hypotheses are true, and the published record is approximately representative of all tested hypotheses. Target state for a healthy field.
High ERR, Low EDR	A research program where published effects are real and well-powered, but there is massive publication bias — many hypotheses are being tested and only the successful ones appear in print. The significant findings will replicate, but the published literature is a heavily filtered view of an extremely low base-rate testing environment.
Low ERR, High EDR	A research program where most tested hypotheses are true but studies are systematically underpowered. Published effects are real but fragile — they will often fail to replicate because the original studies were just barely significant. The solution is larger sample sizes, not better hypotheses.
Low ERR, Low EDR	The worst-case scenario: underpowered studies, low true positive rate, and likely substantial p-hacking to produce publishable results. This pattern characterized the replication crisis in social psychology — a low base rate of true effects combined with underpowered flexible analyses.

4. Example Application: Social Psychology and the Replication Crisis

Bartos and Schimmack (2022) applied z-curve 2.0 to a sample of test statistics from articles published in leading social psychology journals between 2010 and 2017. The z-curve plot showed a pronounced spike at the significance threshold — consistent with widespread p-hacking — and a rapid falloff of z-scores just above the threshold, consistent with publication suppression of non-significant results.

The fitted mixture model estimated an ERR of approximately 54% — meaning that if direct replications were run for all the significant findings in this sample, approximately half would replicate at $p < .05$. This aligns strikingly well with the empirical replication rate observed in the Open Science Collaboration's Reproducibility Project (2015), which found that approximately 36–50% of 100 psychology findings replicated. z-curve's estimate, derived entirely from the shape of the z-score distribution without running any replications, correctly predicted the empirical outcome.

The EDR estimate was substantially lower — approximately 31% — indicating that across all hypotheses being tested (including the non-significant ones that did not appear in print), only about a third were actually true. The gap between the ODR (the fraction of published results that were significant, approximately 95%) and the EDR (31%) quantifies the publication bias in the literature: a 64-percentage-point gap between what appeared in print and the estimated true discovery rate.

Subsequent z-curve analyses have been applied to clinical psychology (Fries & Frankenbach 2020), cognitive neuroscience, organizational behavior, management research, and economics. ERR estimates vary substantially across fields, from approximately 40% in some areas of social psychology to approximately 70–75% in cognitive psychology and experimental economics, consistent with known differences in study design and hypothesis testing practices across disciplines.

5. Requirements & Applicability

z-curve 2.0 is computationally and statistically more demanding than the simpler p-value distribution methods, but its requirements are manageable for any researcher with access to a bibliometric database and the R package zcurve.

Test statistics and df	The exact test statistic (t, F, z, r, chi-squared) and degrees of freedom for each reported hypothesis test. z-curve requires exact z-score conversion, which is only possible with the underlying statistic and df. Reported inequalities ($p < .05$) cannot be used and must be excluded.
Both significant and non-significant results	Unlike p-curve and p-uniform, z-curve benefits substantially from including non-significant results. The EDR estimate in particular is highly sensitive to the ratio of significant to non-significant results, which requires a sample that includes both. Literatures where non-significant results are systematically absent (severe file-drawer problem) will produce biased EDR estimates.
Large sample of test statistics	The mixture model fitting is unstable with fewer than approximately 100 z-scores. Reliable ERR and EDR estimates with narrow confidence intervals typically require 200 or more test statistics. z-curve is a field-level tool, not a study-level or small-literature tool.
Consistent test type	All test statistics should correspond to hypothesis tests of the same general type — directional tests of individual effects, not omnibus F-tests or goodness-of-fit statistics. Mixing qualitatively different test types distorts the z-score distribution and invalidates the mixture model.
Focal versus all tests	A critical analyst decision: using only focal hypothesis tests (one per study) versus all reported tests. Using all tests inflates the sample but includes exploratory and secondary analyses that may have different power characteristics than primary tests. The choice should be documented and sensitivity analyses conducted across both approaches.
Raw data required?	No. z-curve operates on reported test statistics. Automated extraction tools (statcheck, custom parsers) can build the z-score dataset from large literature samples. The zcurve R package handles all conversions and model fitting internally.

6. Limitations

- The EDR estimate is highly sensitive to the inclusion or exclusion of non-significant results. If non-significant results are systematically absent from the dataset (severe publication bias), the EDR will be overestimated — the model has no information about the null-result studies and must infer their frequency from the shape of the significant tail alone. Researchers should assess the completeness of their non-significant result coverage before interpreting EDR.

- The mixture model requires choosing K , the number of power components. z-curve 2.0 uses $K = 7$ components by default, which provides sufficient flexibility for most distributions without overfitting. However, the choice of K can affect estimates in edge cases. Sensitivity analyses across different K values are advisable when estimates are used for high-stakes conclusions.
- z-curve cannot diagnose the source of low power or low EDR. A low ERR could reflect underpowered studies, measurement error, researcher degrees of freedom, or genuine null effects mixed into a literature testing both true and false hypotheses. The tool diagnoses the statistical signature; interpreting what produced it requires domain knowledge.
- The tool is not appropriate for literatures with heterogeneous test types or mixed directional and non-directional hypotheses. Mixing omnibus tests, interaction terms, and simple main effects creates a z-score distribution that does not conform to the mixture model's assumptions.
- z-curve estimates come with substantial uncertainty. Confidence intervals on ERR and EDR are typically wide — 95% CIs of ± 10 –15 percentage points are common with samples of 200 z-scores. Point estimates should always be reported with their confidence intervals. Comparing ERR estimates across fields requires overlapping confidence intervals before concluding the fields differ.
- The model assumes that selection into the published record is determined solely by significance at $p < .05$. In practice, selection is more complex — effect size, sample size, novelty, and journal prestige also affect publication probability. Violations of the simple selection assumption bias the EDR estimate in ways that are difficult to assess.

7. How to Apply

Building the z-score dataset

Define inclusion criteria before extracting statistics. Use statcheck or a custom parser to extract test statistics from PDF files, or manually code statistics from a random sample of papers in the target literature. For each test, record the statistic type, value, and degrees of freedom. Convert all to z-scores using the zcurve R package's conversion utilities or the qnorm/pt functions directly.

Running z-curve in R

- Install and load the package: `install.packages('zcurve'); library(zcurve)`.
- Create the z-score vector: `z <- c(2.1, 3.4, 1.8, 2.9, ...)` or pass p-values directly via `z <- qnorm(1 - p/2)` for two-tailed tests.
- Fit the model: `fit <- zcurve(z)`. The default uses $K = 7$ components with the EM algorithm.
- Extract estimates: `summary(fit)` returns ERR, EDR, and their 95% bootstrap confidence intervals. `plot(fit)` produces the z-curve plot with the fitted mixture overlay.
- Run sensitivity analyses: `refit <- zcurve(z, method = 'density')` for the kernel density alternative, and compare ERR and EDR estimates across methods.

Reporting standards

Report the number of z-scores analyzed, the focal test selection rule, the ERR and EDR estimates with 95% confidence intervals, the observed discovery rate (ODR) for comparison, and the z-curve plot. State explicitly whether significant and non-significant results were both included, and if non-significant results were excluded, note that EDR estimates will be upwardly biased. Compare ERR to any available empirical replication rate data as a validation check.

8. z-curve 1.0 vs. z-curve 2.0

z-curve 1.0 (Schimmack & Brunner 2017) estimated only the ERR — it fitted the mixture model to significant z-scores only, producing an estimate of expected replication rate without an EDR. z-curve 2.0 extends the approach in two important ways: it incorporates non-significant z-scores into the likelihood, substantially improving the stability of the power mixture fit; and it derives the EDR from the fitted mixture and the observed discovery rate, adding a publication bias estimate that was entirely absent from the original version. In practice, z-curve 2.0 should be preferred in all applications — it is strictly more informative than 1.0 and the R package implements both for backward compatibility.

9. Related Tools

p-curve	Operates only on significant p-values; diagnoses evidential value and estimates average power without modeling the full z-score distribution. p-curve is more appropriate for small literature samples (20–50 studies) where z-curve's mixture model would be unstable. Run p-curve for evidential value, z-curve for field-level replication rate estimation.
p-uniform / p-uniform*	Produces a bias-corrected effect size estimate alongside an evidential value test; also operates only on significant results. Complementary to z-curve: use p-uniform for a bias-corrected effect size, z-curve for a bias-corrected replication rate and discovery rate.
Caliper test	Tests for a spike at the significance threshold; a simple diagnostic of threshold-seeking behavior. The caliper test's positive signal is consistent with the spike near $z = 1.96$ visible in z-curve plots of p-hacked literatures. The two tools are complementary diagnostics of the same underlying problem.
Excess Significance Test (TES)	Compares observed to expected significant result counts given study power. z-curve and TES both estimate expected power but use different methods: z-curve from the full z-score distribution, TES from assumed effect sizes and sample sizes. Convergence of z-curve's low ERR with a positive TES provides strong evidence of systematic reporting problems.
Open Science Collaboration (2015)	The empirical benchmark against which z-curve ERR estimates can be validated: 36–50% replication rate in psychology. z-curve's estimates for the same body of literature closely predicted this rate, providing the method's most important external validation.

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THE GARDEN OF FORKING PATHS

Multiverse
Analysis ·
Specification
Curve Analysis ·
Vibration of
Effects

At a glance: These three tools share a common purpose: making visible how much a published finding depends on analytic choices the researcher made — choices that are rarely disclosed, often arbitrary, and collectively capable of turning a null result into a significant one, or vice versa.

1. The Shared Problem: Analytical Fragility

Every empirical study involves dozens of decisions that are never fully pre-specified: which cases to exclude, which covariates to control for, how to operationalize the key variables, which model family to use, what to do with outliers. Taken individually, each choice seems reasonable. But taken collectively they form what Andrew Gelman and Eric Loken described in 2014 as a 'garden of forking paths': a branching decision tree where the researcher, consciously or not, tends to follow the path that yields significance.

The problem is not that researchers are dishonest. It is that the garden is invisible in the published paper. Readers see one path — the one that reached significance — with no indication of how many equally valid paths were never reported, or what they would have shown. This selective presentation inflates the false-positive rate in the literature without any individual act of fraud.

The three tools described in this guide all operate by making the rest of the garden visible. They differ in how they do this, what data they require, and what quantity they ultimately estimate.

2. At-a-Glance Comparison

Core question	Which results are possible across all defensible specifications? What is the distribution across all valid specs with inferential tests? How sensitive is the effect to covariate choice?
Primary output	Distribution of effect sizes across all specification combinations
Disciplines	Psychology, sociology, political science; economics (SCA); epidemiology (VoE)
Raw data needed?	Yes — all three require individual-level data to re-run analyses from scratch
Introduced	Multiverse: Steegen et al. 2016 · Specification Curve: Simonsohn et al. 2020 · Vibration of Effects: Patel et al. 2015

Tool A · Multiverse Analysis

Running all defensible specifications and displaying the full distribution of results

Overview

Multiverse analysis, introduced by Sara Steegen and colleagues in 2016, takes the position that there is rarely a single correct specification of a statistical model — there is instead a set of equally defensible choices. Rather than choosing one and reporting it as 'the' result, multiverse analysis runs all of them and presents the entire distribution of effect sizes and significance decisions as the finding.

Methodological Overview

The analyst begins by enumerating every analytic decision point where multiple choices would be scientifically defensible. Common decision nodes include:

- Outlier exclusion rules (e.g., ± 2 SD vs. ± 3 SD vs. no exclusion)
- Variable operationalization (e.g., scale total vs. mean; raw vs. log-transformed)
- Covariate inclusion (e.g., include age, exclude age, include age and gender)
- Subsample restrictions (e.g., full sample vs. participants meeting an attention check)
- Model family (e.g., OLS vs. robust regression vs. mixed effects)

Each unique combination of one choice at each node constitutes a 'specification'. If there are 3 outlier rules, 2 operationalization, 3 covariate sets, and 2 subsamples, there are $3 \times 2 \times 3 \times 2 = 36$ specifications. The

multiverse runs all 36 and records the effect size and p-value for each. Results are displayed as a two-panel figure: effect size estimates sorted by magnitude (top panel), and a grid showing which analytic choices produced which estimates (bottom panel).

Example Application

Oswald et al. conducted a multiverse-style re-analysis of studies on whether implicit racial bias (IAT scores) predicted discriminatory behavior. Using different inclusion criteria, moderator sets, and outcome operationalizations, the distribution of effect sizes across specifications ranged from near-zero to substantial — with the original published estimate sitting at the positive tail of the plausible range, not the center. The multiverse made visible that the headline estimate was among the most favorable of all defensible results.

Requirements

Raw data	Required. The multiverse re-runs the analysis from scratch under each specification. Published summary statistics alone are not sufficient.
Code access	Highly beneficial. Reproducible analysis code (R, Python, Stata) allows automated re-running. Without code, each specification must be run manually.
Decision enumeration	The researcher must explicitly specify all choice nodes and defensible options at each. This requires domain knowledge — there is no automated way to enumerate valid choices.
R package	The multiverse R package (Steegeen et al.) automates specification enumeration and running. <code>specr</code> is an alternative with a tidy interface.
Interpretive rule	Results require a decision rule for 'robust': proportion of significant specs, sign consistency, or the width of the effect size distribution across the full multiverse.

Tool B · Specification Curve Analysis

Plotting all valid specifications with inferential permutation tests across the full curve

Overview

Specification curve analysis (SCA), developed by Uri Simonsohn, Joseph Simmons, and Leif Nelson, takes a similar approach to multiverse analysis but adds a formal inferential layer: instead of just displaying the distribution, it tests whether the observed curve is more extreme than would be expected if the null hypothesis were true across all specifications.

Where multiverse analysis asks 'what do the results look like across the choice space?', SCA asks 'is there genuine signal in the data, or is the pattern of results consistent with chance given the number of choices made?' The specification curve is simultaneously a visualization tool and a statistical test.

Methodological Overview

SCA proceeds in three steps. First, define the universe of specifications — all combinations of analytic choices that are (a) not clearly wrong and (b) at least defensible given the research question. Second, estimate and plot the specification curve: each specification is run, its effect and significance recorded, results sorted by effect size and plotted as a curve. Third, inferential tests:

Test 1: Median β across all specs vs. permutation null

Test 2: % specs significant vs. chance expectation

Both tests use permutation: the outcome variable is randomly shuffled many times, the full specification curve is estimated for each shuffle, and the observed curve statistics are compared to the null distribution.

A finding is considered robust if (a) the median effect size across all specifications is significantly larger than the permutation null, and (b) the proportion of significant specifications is significantly higher than expected by chance.

Example Application

Orben and Przybylski (2019) applied SCA to whether screen time harms adolescent wellbeing. Using three large longitudinal datasets, they enumerated 43,890 specifications arising from different wellbeing outcomes, screen time measures, covariate sets, and subsamples. The specification curve showed that the vast majority of effect sizes were negative but extremely small — comparable in magnitude to wearing glasses. The inferential tests confirmed that while the association was real, it was far too small to support public discourse claims. The SCA replaced a single contested estimate with a complete picture of the effect landscape.

Requirements

Raw data	Required. SCA re-estimates the model under every specification from scratch.
Reproducible code	Strongly recommended. The <code>specr</code> R package and <code>rdanalysis</code> automate specification enumeration and curve estimation.
Pre-specification	Most credible when the universe of valid specifications is defined before looking at results — otherwise the choice of 'valid' specs can itself become a researcher degree of freedom.
Sample size / spec count	The permutation test gains power with either a large sample or many specifications. With very few specs and a small sample the test may have low power even when a genuine effect exists.

Tool C · **Vibration of Effects (VoE)**

Quantifying sensitivity of epidemiological associations to covariate perturbations

Overview

Vibration of effects, introduced by Chirag Patel and colleagues in 2015, addresses a specific and pervasive form of analytical fragility in epidemiology and clinical research: the sensitivity of a reported association to the choice of adjustment covariates. In fields where researchers routinely control for 'relevant confounders', the choice of which confounders to include is often undisclosed and arbitrary — and it can dramatically change both the magnitude and direction of the reported association.

Methodological Overview

The analyst begins with a set of k candidate covariates. VoE fits the regression model for every possible subset of those covariates, producing 2^k models in total. For each model, the effect size estimate and its confidence interval are recorded. The resulting distribution is summarized by three quantities:

VoE ratio = $\max(|\beta|) / \min(|\beta|)$ across all covariate subsets
Sign instability = % of models where β changes sign
CI overlap = % of models where CI excludes zero

A VoE ratio near 1 indicates a robust association. A ratio > 2 indicates substantial covariate sensitivity. Sign instability > 0% is particularly alarming — it means the direction of the association depends on which covariates are included.

Example Application

Patel et al. applied VoE to 13 associations between common exposures and health outcomes published in high-impact journals. Most associations showed substantial vibration — some changed sign depending on covariate choice, and many had VoE ratios above 2. The authors concluded that many epidemiological findings are better understood as covariate-dependent rather than reflecting stable causal relationships. The coffee-and-mortality literature is a classic example: studies that control for smoking tend to find a protective effect of coffee; those that do not tend to find null or harmful effects. VoE quantifies this instability formally.

Requirements

Individual-level dataset	Required. VoE needs access to raw data to re-fit regression models with different covariate subsets. Published summary statistics are insufficient.
Candidate covariate list	A pre-defined set of plausible confounders. Most credible when the candidate set is specified a priori or drawn from a domain consensus list.
Computational resources	For k candidate covariates, VoE fits 2^k models. For $k = 10$ this is 1,024 models — manageable. For $k = 20$ it is over one million, requiring efficient computing. The <code>voe</code> R package handles this.
Consistent outcome & exposure	All models must use the same outcome variable and exposure variable; only the covariate set varies. If outcome or exposure also vary, the tool shades into multiverse territory.

6. Shared Limitations

- All three tools require access to individual-level raw data. They cannot be applied post-publication without data sharing by the original authors — making them primarily useful as prospective methods or in re-analysis contexts where data are available.
- The definition of 'valid' or 'defensible' specifications is itself a researcher choice. Pre-registration of the specification universe is the only guard against this becoming a new researcher degree of freedom.
- A finding that is fragile across the specification space is not necessarily wrong. There may be theoretical reasons to prefer one specification over others. These tools diagnose instability; they do not determine the truth.
- Computational burden scales quickly. Multiverse and SCA become expensive for large datasets with many choice nodes; VoE becomes expensive for large covariate sets. Published R packages (`multiverse`, `sprc`, `voe`) mitigate but do not eliminate this.
- These tools are not primarily fraud-detection tools — they detect fragility, which is usually the product of normal researcher behavior rather than misconduct. Their primary value is epistemic: they tell us how confident we should be in a finding.

7. Related Tools

p-curve	Tests whether a literature shows genuine evidential value; can detect p-hacking at the study-set level but does not reveal which analytic choices produced the significant results.
z-curve 2.0	Estimates the expected replication rate from published z-values; complements specification curve by quantifying how much of a literature's signal is likely to survive replication.
SPRITE / GRIM	Operate at the single-study level on reported summary statistics; complementary to the three tools here, which operate on the raw analytic pipeline.
Preregistration	The prospective counterpart: committing to a single specification before data collection removes researcher degrees of freedom at the source rather than diagnosing them after the fact.

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**CORRECTING
FOR THE FILE
DRAWER**

**Tier 2a / 3 ::
Publication Bias**

At a glance: These four tools all address the same distortion: because journals preferentially publish significant positive results, the literature that reaches readers is not a representative sample of all studies conducted. Each tool uses different mathematical logic to detect or correct for this bias in a body of published evidence.

1. The Shared Problem: The File Drawer

Publication bias operates at two levels simultaneously. At the team level, researchers file away null results and submit only significant findings. At the journal level, editors preferentially accept papers reporting significant or novel outcomes. The combined effect is that the published record systematically overstates both the prevalence and the magnitude of true effects.

The consequences are significant. Meta-analyses — which are supposed to synthesize the best available evidence — absorb this distortion directly: if the input studies are biased toward significance, the pooled estimate will be inflated, sometimes dramatically. Clinical guidelines, policy decisions, and further research programs are then built on inflated foundations.

Detecting and correcting for publication bias requires indirect inference — we are trying to estimate the shape of a distribution from a visible, biased sample of it. The four tools here take different approaches to that inference problem, with different assumptions, different outputs, and different sensitivities to violations of those assumptions:

Dimension	Funnel Plot / Egger Test	Trim-and-Fill	PET-PEESE	Excess Significance (TES)
Core logic	Asymmetry in a plot of effect size vs. precision	Impute missing studies to restore funnel symmetry	Model the relationship between effect size and SE to separate signal from bias	Compare observed to expected number of significant results
Primary output	Visual + Egger's regression test statistic	Bias-corrected pooled effect estimate	Bias-corrected true effect (PET = 0 test, PEESE estimate)	χ^2 statistic; excess significant findings count
Raw data?	No — published effect sizes and SEs	No — published effect sizes and SEs	No — published effect sizes and SEs	No — published effect sizes and power estimates
Introduced	Light (1984); Egger et al. (1997)	Duval & Tweedie (2000)	PET: Stanley & Doucouliagos (2014); PEESE: Stanley (2017)	Ioannidis & Trikalinos (2007)
Best used for	Initial visual screening; Egger adds inferential rigor	Bias-corrected effect after asymmetry confirmed	Literatures with many studies and varying precision	Literatures where power can be estimated a priori

Tool A · Funnel Plot & Egger Test

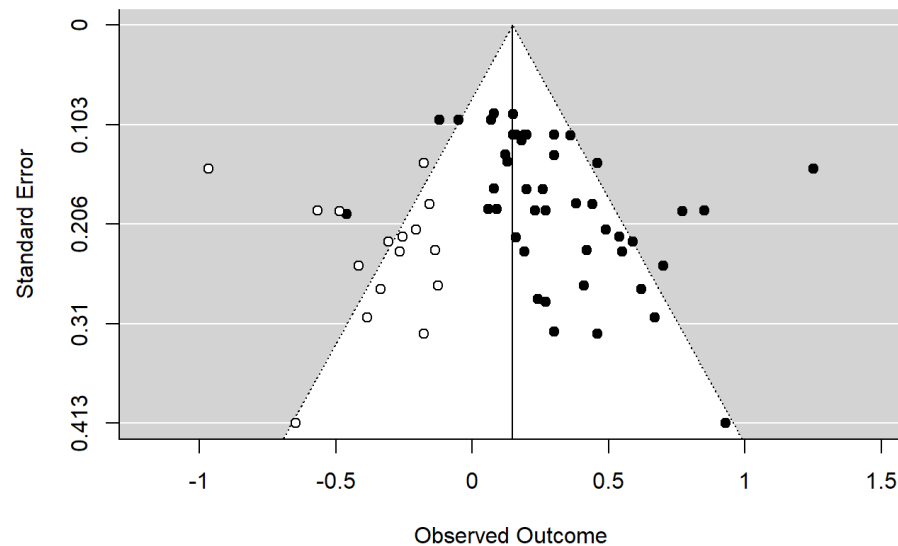
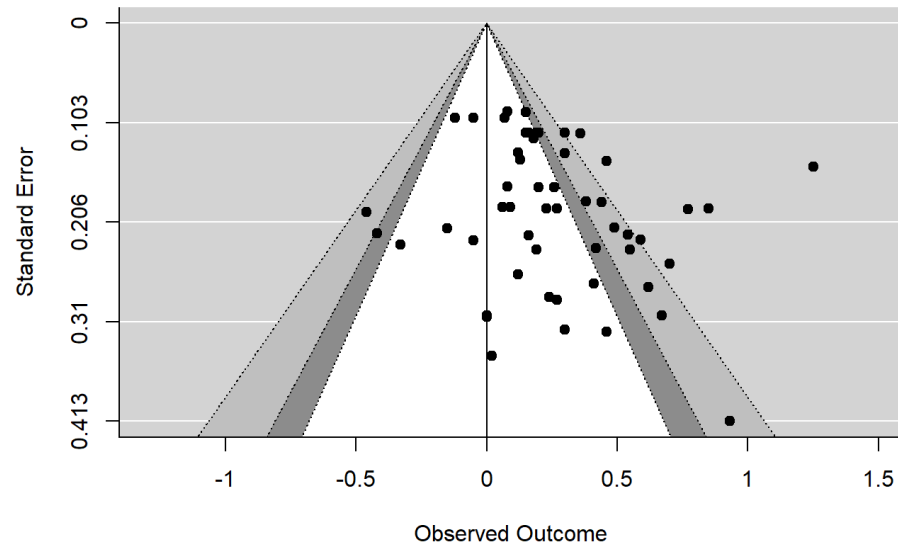
Visual detection of publication bias through asymmetry in effect size distributions

Overview

The funnel plot is the foundational diagnostic for publication bias in meta-analysis. It works from a simple observation: in the absence of bias, the relationship between a study's effect size and its precision (the inverse of its standard error) should be symmetric. Large, precise studies cluster near the true effect; small, imprecise studies fan out around it on both sides equally. The resulting shape is an inverted funnel.

When publication bias is present, the bottom-left of the funnel — small studies with small or null effects — is missing. Small studies that found nothing were never published; only the small studies that happened to find large effects made it into print. The funnel becomes asymmetric, with a gap at the bottom-left and an excess of studies at the bottom-right.

The Egger test, introduced in 1997, converts the visual inspection into a formal statistical test. It regresses the standardized effect size on precision and tests whether the intercept is significantly different from zero. A non-zero intercept indicates funnel asymmetry — which may reflect publication bias, but could also arise from genuine between-study heterogeneity, poor methodological quality in small studies, or artifactual correlations between sample size and effect size.



Example funnel plot and funnel plot with imputed missing studies added.

From: <https://civanlissa.github.io/Doing-Meta-Analysis-in-R/smallstudyeffects.html>

Constructing a funnel plot requires three quantities for each study in the meta-analysis: the effect size estimate (d, r, OR, etc.), its standard error (SE), and optionally the sample size (used as an alternative precision axis). The plot displays effect size on the x-axis and precision (1/SE) or sample size on the y-axis, with each study as a point and a reference line at the pooled effect estimate.

The Egger test formalizes the asymmetry check:

$$\text{Standardized effect} = a + b \times (1/\text{SE})$$

Where *a* is the intercept and *b* is the slope. Under no bias: *a* = 0.

H₀: a = 0 (symmetric funnel, no bias)

A significant intercept (*p* < .10 by convention) is taken as evidence of asymmetry.

Note that the Egger test uses *p* < .10 as the conventional threshold rather than .05, because it is typically underpowered in small meta-analyses. A non-significant Egger test does not rule out publication bias — it may simply reflect insufficient studies to detect the asymmetry.

Example: SSRIs and Depression

Turner et al. (2008, NEJM) used funnel plot inspection alongside registry analysis to demonstrate that the published literature on SSRIs dramatically overstated efficacy. Of 74 FDA-registered trials, 38 with positive results were published; 37 with negative or questionable results were either unpublished or published in a way that conveyed a positive outcome. The funnel plot of published studies was visibly asymmetric. When the unpublished trials were included, the effect size for SSRIs dropped substantially — from a magnitude that appeared clinically meaningful to one that was borderline. This analysis influenced regulatory guidance on pharmaceutical trial reporting.

Requirements

Effect sizes	The standardized effect size (Cohen's d, correlation <i>r</i> , log odds ratio) and its standard error for each study in the meta-analysis. These are typically available from published papers.
Minimum study count	Funnel plot asymmetry is only interpretable when there are at least 10 studies. Egger's test has very low power below this threshold and should not be reported as a definitive test with fewer than 10–15 studies.
Effect size homogeneity	The funnel plot assumes all studies are estimating the same underlying effect. When genuine heterogeneity exists (different populations, interventions, or outcomes), funnel asymmetry may reflect heterogeneity rather than bias.
No raw data required	The test operates on published summary statistics: effect size and SE per study. Individual-level participant data are not needed.
Contour-enhanced funnel	A useful extension adds contour lines at <i>p</i> = .05, .01, .001 to the funnel plot. If missing studies cluster in non-significant regions, publication bias is a plausible cause; if they cluster in significant regions, heterogeneity is more likely.

Tool B · Trim-and-Fill

Imputing missing studies to restore funnel symmetry and re-estimate the true effect

Overview

Trim-and-fill, developed by Sue Duval and Richard Tweedie in 2000, takes funnel plot logic one step further. Rather than simply detecting asymmetry, it estimates how many studies are missing from the publication record and imputes them — mirroring the observed asymmetric studies to restore symmetry — and then re-calculates a bias-corrected pooled effect estimate with these imputed studies included.

The method gives a concrete answer to the question: 'If all the missing small-N null studies had been published, what would the meta-analytic estimate be?' This makes it one of the most intuitive and widely reported correction methods, though it rests on assumptions that are often violated in practice.

Methodological Overview

The algorithm proceeds iteratively in two phases:

Phase 1 (Trim): Starting with the full set of studies, the method identifies and temporarily removes the most extreme studies on the side of the funnel showing excess (typically the right side, where significant small-N studies cluster). Studies are trimmed iteratively until the funnel is approximately symmetric around the pooled estimate. The number of trimmed studies is the estimated number of missing studies.

Phase 2 (Fill): Mirror images of the trimmed studies are created on the opposite side of the funnel. These imputed studies, combined with the original studies, produce a symmetric funnel. The pooled effect is then re-estimated using all studies — real and imputed.

k_0 = estimated number of missing studies
Estimated from the number of studies that must be trimmed to achieve funnel symmetry.

δ_{adjusted} = pooled effect with k_0 imputed studies added
The bias-corrected estimate. The difference between δ_{adjusted} and the original estimate quantifies the impact of publication bias.

The R_0 estimator (Duval & Tweedie 2000) and the L_0 estimator are the two most common variants; they differ in how they locate the center of the funnel. Results can differ meaningfully between estimators, so sensitivity analyses should report both.

Example: Antidepressants and Placebos

Kirsch et al. (2008) applied trim-and-fill alongside registry data in their influential analysis of SSRI versus placebo effect sizes. Trim-and-fill indicated that several small-N null studies were likely missing from the published record. When these studies were imputed, the bias-corrected effect size for antidepressants fell below the NICE clinical significance threshold for all but the most severely depressed patients. The analysis triggered a public debate about the clinical utility of SSRIs for mild-to-moderate depression and subsequently influenced prescribing guidelines in the US and UK.

Requirements

Published effect sizes and SEs	As with the funnel plot, trim-and-fill requires only the summary statistics (effect size, SE) from each included study — no individual-level data needed.
Sufficient number of studies	The algorithm is unreliable with fewer than 10 studies and should not be applied below this threshold. With very few studies, the trim-and-fill estimate is highly sensitive to the choice of estimator variant.
Symmetric bias assumption	The method assumes that missing studies are exact mirror images of the asymmetric observed studies — a strong assumption rarely fully satisfied. In heterogeneous literatures, trimmed studies may differ qualitatively, not just directionally.
Homogeneous true effect	Trim-and-fill was derived under a fixed-effect model assumption. Under genuine heterogeneity, the method may over-correct (imputing studies that do not exist) or under-correct, and the bias-corrected estimate should be interpreted cautiously.
Complement, not replace	Trim-and-fill results are most credible when they converge with other methods (Egger test, PET-PEESE). Divergent results across methods signal model sensitivity and should prompt explicit sensitivity analysis.

Tool C · PET-PEESE

Regression-based separation of true effect signal from precision-dependent bias

Overview

PET-PEESE (Precision Effect Test – Precision Effect Estimate with Standard Error) is a meta-regression approach developed by Tom Stanley and colleagues that takes a fundamentally different view of publication bias. Rather than trying to identify missing studies and impute them, it models the mathematical relationship between an effect size and its standard error across all published studies, and uses that model to extrapolate what the effect size would be at infinite precision — that is, what signal remains after the bias is removed.

The key insight is that publication bias creates a predictable correlation between effect size and standard error: studies with large standard errors (small samples, imprecise) tend to report larger effects, because small imprecise studies only make it into print when they happen to find something large. PET-PEESE models this correlation and uses it to estimate the bias-free effect.

Methodological Overview

The method proceeds in two sequential steps:

Step 1 — PET (Precision Effect Test): Regress the effect size on the standard error with study-level weights ($1/SE^2$). Test whether the intercept — the predicted effect size when $SE = 0$, i.e., at infinite sample size — is significantly different from zero.

$$\text{Effect size} = \beta_0 + \beta_1 \times SE + \varepsilon$$

Under publication bias: $\beta_1 > 0$ (larger $SE \rightarrow$ larger reported effect). Under no bias: $\beta_1 = 0$.

$H_0: \beta_0 = 0$ (the true effect is zero after correcting for bias)

If PET fails to reject H_0 , the literature may be entirely explained by bias. If it rejects, proceed to PEESE.

Step 2 — PEESE (Precision Effect Estimate with Standard Error): If PET rejects the null (suggesting a non-zero true effect exists), switch to a model that regresses effect size on SE^2 rather than SE . This provides a less biased estimate of the true effect size, because squaring the SE reduces the leverage of the most extreme outliers.

$$\text{Effect size} = \beta_0 + \beta_1 \times SE^2 + \varepsilon$$

The intercept β_0 is the PEESE estimate of the true effect — the predicted effect at $SE = 0$.

The two-step logic is important: if PET suggests zero true effect, reporting the PEESE estimate can be misleading (because PEESE extrapolates assuming a non-zero effect exists). PET acts as a gatekeeper.

Example: Ego Depletion

Carter et al. (2015) applied PET-PEESE to the ego depletion literature — the highly influential finding that acts of self-control deplete a limited mental resource, making subsequent self-control harder. The published meta-analytic effect size was $d \approx 0.62$. PET found a significant relationship between effect size and SE , indicating bias. The PEESE-corrected estimate was $d \approx 0.24$ — less than half the published estimate. A subsequent pre-registered multi-lab replication (Hagger et al., 2016, 23 labs) found $d \approx 0.04$, essentially zero. The PET-PEESE estimate, while not perfectly accurate, correctly identified that the published literature dramatically overstated the true effect.

Requirements

Effect sizes and SEs	Published effect sizes and standard errors for each study in the meta-analysis. Unlike trim-and-fill, PET-PEESE is also valid under heterogeneous random-effects models if the between-study variance is accounted for in the weights.
Minimum study count	Reliable regression requires at least 20–30 studies. With fewer studies, the regression is unstable and the intercept estimate has very wide confidence intervals.
No floor/ceiling effects	PET-PEESE assumes an approximately linear relationship between effect size and SE (PET) or SE^2 (PEESE). This assumption can break down at extreme precision values or when effect sizes are bounded (e.g., proportions near 0 or 1).

Domain knowledge	The two-step PET → PEESE decision rule requires the analyst to correctly judge whether PET's rejection of H_0 reflects a real effect or is itself a product of model misspecification. Context matters.
No individual-level data	Operates entirely on published summary statistics. This makes it among the most accessible post-publication bias checks.

Tool D · **Excess Significance Test (TES)**

Comparing the observed rate of significant findings to what statistical power would predict

Overview

The Excess Significance Test (TES), developed by John Ioannidis and Thomas Trikalinos in 2007, approaches publication bias from an entirely different angle. Instead of examining the distribution of effect sizes, it examines the rate of statistical significance across a literature and asks: given the sample sizes and effect sizes in these studies, how many of them should have been statistically significant, if the underlying effect is real and the studies are conducted properly? If the observed number of significant results far exceeds what power calculations would predict, something is wrong — either the studies are p-hacked, or publication bias has suppressed the null results.

Methodological Overview

The test estimates the expected power of each study under a specific assumed true effect size (typically the pooled estimate from the meta-analysis, or a minimum clinically important difference). It then calculates the expected number of significant results given those power values, and compares that to the observed number using a chi-squared or binomial test.

$$E = \sum_i \pi_i$$

Expected number of significant findings, where π_i is the estimated power of study i .

O = observed number of significant findings in the literature

Test: $O \gg E \rightarrow$ excess significance detected

A formal χ^2 test compares O to E. Significant excess implies p-hacking, publication bias, or both.

The critical assumption is the choice of true effect size used to estimate power. Ioannidis and Trikalinos recommend using the effect size from the largest and most precise study in the literature (rather than the inflated pooled estimate), as this provides a less biased benchmark. Sensitivity analyses across a range of assumed true effects are strongly recommended.

Example: Genetic Associations in Epidemiology

Ioannidis and Trikalinos first applied TES to a set of genetic association studies for common diseases — a field notorious for non-replication. Across multiple meta-analyses, they found that the number of statistically significant findings was several times higher than what the observed power of the individual studies would predict. For example, in a set of 25 studies on a particular gene-disease association, the expected number of significant results (given the observed effect size and sample sizes) was approximately 7, but 19 of the 25 studies reported significance. This excess was consistent with either systematic p-hacking within studies or severe publication suppression of null results — likely both.

TES has since been applied to neuroimaging, nutrition, psychology, and economics. Šimkus et al. (2023) found substantial excess significance in prominent criminology literatures. Tenkanen et al. (2023) found it pervasive in dietary intervention trials.

Requirements

Effect sizes and SEs / sample sizes	To estimate power for each study, you need either the SE (from which approximate power can be calculated given an assumed true effect) or the raw sample size and design details.
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Assumed true effect size	The test requires a benchmark true effect. Options: (1) the effect size from the single largest study; (2) the meta-analytic estimate; (3) a minimum clinically important difference. Each yields different power estimates, so sensitivity analysis across assumptions is essential.
Statistical significance decisions	You need the reported p-value or significance status ($p < .05$ or not) for each study. The test counts the number of studies that crossed the threshold.
Homogeneous outcome	TES works best when all studies are testing the same directional hypothesis. Mixed literatures with different outcomes or different effect directions make the expected count hard to interpret.
No individual-level data	Operates entirely on published summary statistics and significance thresholds. This makes it applicable to any meta-analytic dataset.

4. Shared Limitations & When to Use Each

All four methods share a fundamental constraint: they infer the shape of an unobserved distribution from a biased visible sample. This means their conclusions are probabilistic and assumption-dependent, not definitive. Several specific limitations apply across the set:

- Heterogeneity confounds asymmetry. When true effects genuinely vary across studies (different populations, interventions, or contexts), small-study effects that mimic publication bias can arise even without any bias at all. A small study conducted in a high-effect subpopulation will look like a publication-biased small study with a large effect. All four methods struggle to distinguish genuine heterogeneity from bias-driven asymmetry.
- Power is low in small meta-analyses. Egger's test, trim-and-fill, and PET-PEESE are unreliable with fewer than 10–15 studies. TES requires enough studies to make a meaningful comparison between O and E. Null results from small meta-analyses should not be taken as evidence of no bias.
- The methods frequently disagree. It is common for trim-and-fill to indicate substantial bias while PET suggests a non-zero effect, or for TES to detect excess significance while Egger's test is non-significant. Analysts should report all four, acknowledge the divergence, and avoid cherry-picking the method that supports their preferred conclusion.
- None can recover the true effect with certainty. These methods estimate how much the literature is distorted; they do not directly observe the missing studies. The bias-corrected estimates from trim-and-fill and PEESE are themselves uncertain and model-dependent — they are better understood as lower bounds on the true effect than as precise corrections.

When to use which

Funnel plot + Egger: always, as a first visual and inferential screen. Trim-and-fill: when you need a bias-corrected effect estimate and the literature is reasonably homogeneous. PET-PEESE: when you have ≥ 20 studies and want a regression-based correction, especially in economics or psychology. TES: when you want to test whether the rate of significant findings itself is implausibly high given the studies' power.

5. Related Tools

p-curve	Examines the distribution of p-values within the significant range (0–.05) rather than comparing observed to expected significance rates. Right-skew indicates true signal; flat or left-skew indicates p-hacking. Complements TES by diagnosing within-study flexibility rather than between-study publication suppression.
z-curve 2.0	Estimates the expected replication rate and discovery rate from the distribution of all test statistics, significant and non-significant. Provides a field-level summary of evidential value that situates individual meta-analyses within the broader literature.
Multiverse / Specification Curve	If raw data are available, these tools can quantify how the meta-analytic pooled effect itself varies across defensible inclusion and coding decisions — a higher-level form of publication bias correction.
Registered Reports	The prospective remedy: publishing a commitment to report results regardless of outcome removes publication bias at the source, eliminating the need for post-hoc correction.

Key Citations

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BIBLIOMETRIC ANOMALY DETECTION

Tier 2b :: Citation
Rings & Collusion-
Level Misconduct

At a glance: Bibliometric anomaly detection identifies journals, authors, or institutions whose citation and publication patterns deviate significantly from field-level norms — flagging potential citation rings, coercive citation, editorial manipulation, or paper mill activity. It operates on metadata available from academic databases and does not require access to the underlying manuscripts.

1. Overview

Every researcher, journal, and institution leaves a statistical footprint in the academic record: patterns of who cites whom, how often, and in what context. Under ordinary conditions these patterns reflect the genuine intellectual structure of a field — seminal papers are widely cited, recent publications build on recent work, and citation exchange between groups is roughly proportional to intellectual proximity. Bibliometric anomaly detection asks whether observed patterns deviate from what those normal conditions would predict.

The approach has roots in the sociology of science (Merton, de Solla Price) and became practically powerful as large-scale bibliometric databases — Web of Science, Scopus, OpenAlex, Semantic Scholar — made it possible to compute network statistics across millions of papers. What was once a laborious manual exercise can now be automated across entire journal catalogues.

Unlike tools such as GRIM or statcheck, which examine individual papers for impossible statistics, bibliometric anomaly detection operates at the journal, author, or institutional level. It does not identify fabricated data directly — it identifies structural patterns that are inconsistent with legitimate scholarly activity and that warrant further investigation. A flagged journal or author cluster is a starting point for inquiry, not a finding of misconduct.

2. Methodological Overview

Several distinct bibliometric signals are used in combination. No single metric is sufficient on its own — the diagnostic value comes from the convergence of multiple anomalies, or from the severity of a single anomaly relative to a well-calibrated field baseline.

Self-citation rate

Self-citation — authors citing their own prior work — is normal and legitimate up to a point. The anomaly threshold varies by field: in highly specialized niches, rates above 20–25% are common; in broad fields, rates above 10–15% are suspicious. Extreme self-citation (> 50%) can artificially inflate h-indices and impact metrics and is flagged by several databases. The key metric is whether the self-citation rate significantly exceeds the distribution of comparable authors in the same subfield and career stage.

Co-citation and reciprocal citation networks

Citation network analysis maps directed edges between papers. Anomalies of interest include:

- Dense reciprocal citation clusters: groups of authors who disproportionately cite each other relative to their citation of the broader field. Network modularity and clustering coefficients identify these sub-communities.
- Implausible citation velocity: a paper accumulates citations far faster than its intellectual predecessors in the same line of work, or than papers of comparable type from the same period.
- Citation concentration: a journal's impact factor is driven by a small number of papers that are cited extremely heavily relative to the rest of the journal's output — consistent with editorial self-promotion or cite-stacking.

Journal-level metrics

Several journal-level statistics can be computed from database metadata and compared to field distributions:

Self-citation ratio = citations_from_same_journal / total_citations_received

Ratios above 0.3–0.4 are flagged by ESCI and Scopus review committees as potentially manipulated. Clarivate has suppressed impact factors for journals exceeding this threshold.

Co-citation density = intra-cluster citations / total citations (normalized by cluster size)

Compares observed citation density within an author or journal cluster to the expected density under a null model of random citation with the same degree distribution.

Additional journal-level signals include: unusually rapid peer review turnaround times (consistent with fake or superficial review); implausibly high acceptance rates; concentration of authorship in a narrow geographic or institutional cluster; and discordance between stated and actual editorial board membership.

Author-level metrics

For individual authors, anomaly detection examines:

- h-index inflation: an h-index that substantially exceeds what would be expected given the author's publication record, career length, and subfield, suggesting inflated citation through a network.
- Productivity outliers: paper output rates that significantly exceed field norms and are not explained by large-team or consortium authorship. Paper mills generate authorship credits at implausible rates.
- Authorship network anomalies: an author appears across many papers in fields with no apparent intellectual connection to their stated expertise — a signature of purchased or gifted authorship.

3. Example Application: Journal Impact Factor Manipulation

Clarivate Analytics has suppressed the Journal Citation Reports impact factor for multiple journals following bibliometric audit. A well-documented case involved a cluster of journals in the cell biology and chemistry space that engaged in coordinated cite-stacking: editors solicited or required authors to cite a specific set of papers from the same journal as a condition of acceptance.

The bibliometric signature was distinctive: the journals' self-citation ratios were between 0.35 and 0.60 — far above the field baseline of 0.08–0.12 — and the cited papers were concentrated in a small number of issues rather than distributed across the journal's history. Network analysis showed that the citing papers were often methodologically unrelated to the cited papers, confirming the citations were instrumental rather than substantive.

Clarivate suppressed the impact factors of the implicated journals, removing them from the Journal Citation Reports for two years. Several were ultimately delisted from the Science Citation Index entirely. The case illustrated both the power of bibliometric anomaly detection — the pattern was identifiable without reading any individual paper — and its limitation: some journals engaged in coordinated self-citation for years before the statistical signal crossed the detection threshold.

A second class of examples involves paper mills. Richardson & Amaral (PNAS 2025) applied network analysis to PLOS ONE editorial and authorship data, finding that 0.25% of editors were responsible for 30.2% of retracted articles. The bibliometric signal — concentration of editorial throughput in a tiny fraction of the editor network, combined with anomalous authorship clustering in those submissions — allowed identification of the implicated editors before full forensic review of the manuscripts themselves.

4. Requirements & Applicability

Bibliometric anomaly detection requires access to structured citation metadata. The quality and completeness of that metadata is the primary constraint on what can be detected.

Citation database access	
	Web of Science, Scopus, or OpenAlex (free) provide the citation metadata needed. OpenAlex is fully open-access via API and covers > 250 million works. Semantic Scholar provides similar coverage. Some analyses require institutional database licences.

Defined comparison baseline	Anomalies are only detectable relative to a baseline. This requires either a reference set of comparable journals or authors in the same subfield, or a statistical null model of expected citation behavior. The choice of baseline is a significant analyst decision.
Network analysis tools	Citation network construction and community detection require tools such as VOSviewer, Gephi, or network analysis packages in R (igraph, tidygraph) or Python (NetworkX). Bibliometrix (R) provides a complete pipeline from database export to network statistics.
Database export format	Web of Science and Scopus export structured reference data in formats that bibliometric tools can ingest directly. OpenAlex provides a REST API. Raw DOI or title lists are insufficient — full reference metadata is required for network construction.
Field-level norms	Interpretation requires knowing what normal self-citation, citation velocity, and network density look like in the target field. These vary substantially — high in mathematics and physics, lower in biomedical research. Anomalies must be evaluated against field-specific distributions, not universal thresholds.
Time window definition	Citation accumulation is time-dependent. Analyses must specify a publication year window and a citation observation period. Comparing citation rates across different time windows produces spurious anomalies.
No manuscript access required	The method operates entirely on citation and authorship metadata. Full text of the papers is not needed, which makes it scalable to large literature sets.

5. Limitations

- Bibliometric anomalies are consistent with misconduct but not diagnostic of it. A dense citation cluster may reflect a legitimate intellectual community. High self-citation may reflect a researcher at the frontier of a narrow specialization. Anomaly detection generates hypotheses for further investigation, not findings.
- Field norms vary enormously and are poorly documented for many subfields. An anomaly in one field may be standard practice in another. Applying generic thresholds (e.g., 'self-citation > 30% is suspicious') across fields generates both false positives and false negatives.
- Database coverage is uneven. OpenAlex and Semantic Scholar are rapidly improving but still undercount citations in regional-language journals, conference proceedings (especially outside computer science), and pre-2000 literature. Anomalies in poorly covered fields may be artefacts of missing data.
- Legitimate citation incentives mimic the anomaly signature. Impact factor pressure encourages authors to cite journals they want to publish in, generating organic cite-stacking without editorial coercion. Distinguishing intentional manipulation from rational response to incentives is difficult.
- Network methods depend heavily on the chosen community detection algorithm and resolution parameters. Different algorithms applied to the same citation network can produce very different community boundaries, changing which clusters appear anomalous.
- The method is retrospective and slow. Citation patterns take years to become statistically distinguishable. Rapidly expanding paper mill operations may not trigger anomaly flags until hundreds of fraudulent papers have already been published and cited.

6. How to Apply

Step 1 — Define scope and baseline

Specify the target entity (journal, author, institution) and the field. Export a reference set of comparable journals or authors from the same Web of Science or Scopus category. This reference set forms the empirical baseline for anomaly thresholds.

Step 2 — Export citation metadata

From Web of Science or Scopus, export full reference data (not just abstract records) for the target entity and the reference set. In OpenAlex, use the Works API with filters on source, author, or institution. Retain all metadata fields: DOI, references, authors, affiliations, publication date.

Step 3 — Compute self-citation and journal-level metrics

Using Bibliometrix (R) or the bibliometrix web app (bibliometrix.org), import the exported data and compute: self-citation ratio, h-index, citation distribution, and co-citation matrix. Compare values to the reference set distribution. Flag entities in the top or bottom 5th percentile on multiple metrics simultaneously.

Step 4 — Build and analyze the citation network

Construct a directed citation network where nodes are papers or authors and edges are citations. Use igraph (R) or NetworkX (Python) to compute: modularity, clustering coefficient, betweenness centrality. Apply community detection (Louvain or Leiden algorithm). Visualise with VOSviewer or Gephi. Flag communities with internal edge density substantially higher than expected under a configuration null model.

Step 5 — Cross-reference with retraction and alert databases

Check flagged journals and authors against the Retraction Watch Database, Clarivate's suppressed journal list, and Beall's List of predatory publishers. Convergence of bibliometric anomalies with known misconduct indicators substantially increases the probability of genuine problems.

7. Related Tools

Citation Network Analysis	Specifically maps the directed graph of citation relationships to identify clusters and community structure. Bibliometric anomaly detection draws on citation network methods but adds statistical comparison to field baselines.
Editor Network Analysis	Applies network methods to editorial submission routing rather than citation patterns. Richardson & Amaral (PNAS 2025) used this to identify editors enabling fake peer review at scale — a structural complement to citation-level anomaly detection.
Problematic Paper Screener	Operates at the manuscript level, flagging tortured phrases and image duplication. Convergence between a Problematic Paper Screener flag and a bibliometric anomaly on the same journal provides much stronger evidence of systematic misconduct than either alone.
Hijacked Journal Checker	Identifies journals whose ISSNs have been cloned by scam publishers. A prerequisite check before bibliometric analysis — if a journal is hijacked, its citation metadata may be split across the legitimate and fraudulent versions, distorting metrics.
Retraction Watch Database	Provides the ground truth for calibrating bibliometric anomaly thresholds: journals with high retraction rates can be used as positive controls when building detection models.

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FAIR COMPLIANCE ASSESSMENT

Tier 3 :: Poor Data
Stewardship

At a glance: FAIR compliance assessment tools evaluate whether a research dataset or repository meets the FAIR principles — Findable, Accessible, Interoperable, and Reusable — published by Wilkinson et al. in 2016. Unlike the other tools in this guide, FAIR assessment does not detect misconduct in published papers. Instead, it measures the structural quality of data stewardship: whether data can be found by a machine, retrieved without restriction, understood in another context, and legally reused by other researchers. Poor FAIR compliance is a field-level indicator of systemic data loss and reproducibility risk.

1. Overview

The reproducibility crisis has two distinct dimensions. The detection tools in this guide address the first: statistical and methodological failures in published research. The FAIR principles address the second: the structural failure of research data infrastructure. Even if every paper in a literature were methodologically perfect, a field in which datasets are inaccessible, poorly described, or stored in proprietary formats cannot be reproduced, reanalyzed, or built upon. Data that cannot be found or understood is, for scientific purposes, data that does not exist.

The FAIR Guiding Principles, published in *Scientific Data* by Wilkinson and colleagues in 2016, articulate the properties that data and their associated metadata must have to be genuinely reusable. The principles were designed to be implemented by both humans and machines — recognizing that at the scale of modern science, manual assessment of data quality is not feasible. FAIR compliance assessment tools operationalize the principles into scorable, automatable criteria that can be applied to individual datasets, repository-wide collections, or entire research programs.

FAIR assessment sits at Tier 3 in this framework — it is a field-level tool rather than a paper-level or study-level one. A single dataset failing a FAIR check is a management problem. A systematic pattern of low FAIR compliance across a research domain is a structural integrity problem: it means the field cannot audit its own evidence base, replication is practically impossible even for well-intentioned researchers, and the published literature is built on a foundation that cannot be independently verified.

2. The FAIR Principles in Detail

Each of the four principles — Findable, Accessible, Interoperable, Reusable — is operationalized into a set of sub-principles, each of which can be scored independently. The sub-principles are numbered F1–F4, A1–A1.2, I1–I3, and R1–R1.3 in the original Wilkinson et al. paper. FAIR assessment tools map each sub-principle to a testable criterion.

	Principle	What it requires	Why it matters for detection
F	Findable	Persistent identifier (DOI, ARK, PURL) assigned to both data and metadata. Rich metadata describing the data is indexed in a searchable resource. Metadata explicitly includes the identifier for the data it describes.	Datasets that cannot be found by machines cannot be automatically aggregated for meta-analysis, replication matching, or bibliometric analysis. A literature of unfindable data cannot be systematically reviewed.
A	Accessible	Data and metadata are retrievable via a standardized, open, and free protocol (e.g., HTTP, FTP). Authentication and authorization are permitted but must use open, implementable standards. Metadata remains accessible even if the data itself is no longer available.	Gated or expired data links make computational reproducibility impossible. Accessible metadata-after-data-deletion preserves at least the record of what existed, enabling detection of selective data retention.
I	Interoperable	Data use a formal, accessible, shared, and broadly applicable language for knowledge representation. Metadata use vocabularies that follow FAIR principles. Metadata include qualified references to other data.	Interoperable data can be automatically combined across studies — enabling automated p-curve construction, replication matching, and cross-study consistency checking without manual harmonization.
R	Reusable	Metadata include a plurality of accurate and relevant attributes. Data are released with a clear and accessible data usage license. Metadata meet	Without a clear license, data cannot legally be reused for reanalysis. Without provenance, the chain of custody from raw collection to

		domain-relevant community standards. Data and metadata are associated with their provenance.	published summary statistic cannot be audited.
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3. FAIR Assessment Tools

Several automated and semi-automated tools have been developed to score datasets against the FAIR sub-principles. They differ in their scope, the level of automation they achieve, and the communities for which they were designed.

F-UJI (Automated FAIR Data Assessment Tool)

F-UJI, developed by Devaraju and Huber (2021) at the German National Library of Science and Technology, is a fully automated web service that accepts a dataset's persistent identifier or landing page URL and returns a scored assessment across 17 FAIR sub-principles. It uses a combination of content negotiation, metadata harvesting, and vocabulary lookup to assess each criterion without human intervention.

F-UJI is the most widely used automated FAIR assessment tool and is recommended by the European Open Science Cloud (EOSC) for repository compliance monitoring. It returns a structured JSON result with pass/fail scores for each sub-principle, a maturity level score (0–4) for each FAIR category, and an overall FAIR score. The tool is accessible via a REST API, making it suitable for batch assessment of large repository collections.

Input	A persistent identifier (DOI, Handle, ARK) or the URL of the dataset's landing page.
Output	JSON result with sub-principle scores, maturity levels (0–4) per FAIR category, and an overall compliance summary. A web UI at f-ujl.net provides a human-readable report.
Scope	Assesses the metadata and structural properties of the dataset landing page. Does not evaluate the scientific quality or completeness of the data itself.
Automation	Fully automated. No human judgment required for standard assessments. Can be called programmatically via API for batch evaluation of repository holdings.
Limitations	Relies on publicly accessible metadata. Datasets in repositories that use non-standard metadata formats or that restrict metadata access may score poorly even when FAIR-compliant in practice.

FAIRshake

FAIRshake, developed by the Ma'ayan Laboratory at the Icahn School of Medicine at Mount Sinai, takes a community-driven approach. Rather than a single automated scoring algorithm, FAIRshake provides a framework for defining FAIR rubrics tailored to specific communities — biomedical databases, computational tools, clinical datasets — and allows both human reviewers and automated scripts to apply those rubrics.

FAIRshake was used extensively in the NIH Data Commons pilot program and has been applied to assess compliance across large biomedical data repositories including GEO, dbGaP, and the Metabolomics Workbench. Its community-specific rubrics make it more sensitive to domain norms than generic tools, at the cost of requiring rubric development effort before application.

Input	Dataset URL, identifier, or repository API endpoint. Community-specific rubric must be selected or defined before scoring.
Output	Per-criterion scores, visual FAIR insignia (a standardized badge system), and aggregate compliance statistics across a collection.
Scope	Can assess individual datasets, tools, or entire repository collections. Particularly well-suited for community-specific compliance monitoring.
Automation	Partially automated. Core metadata checks are automated; some rubric criteria require human judgment. Batch API access available.

Limitations	Requires community-specific rubric development. Results are not directly comparable across communities using different rubrics.
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FAIR Evaluator

The FAIR Evaluator, maintained by the GO FAIR initiative and accessible at evaluator.fairdata.org, focuses on machine-actionable FAIR — specifically testing whether metadata is structured in a way that a software agent could discover, retrieve, and interpret the dataset without human assistance. It applies a set of FAIR Maturity Indicators, each of which is tested by a specific automated compliance test.

The FAIR Evaluator is particularly relevant for precision medicine and biomedical research, where the goal is not just that a human researcher can find and use data, but that computational pipelines can automatically integrate data across studies. It places heavier weight on semantic interoperability — whether metadata uses controlled vocabularies and ontologies that machines can reason about — than general-purpose tools like F-UJI.

4. Example Application: NIH-Funded Research Repositories

Gabelica et al. (2022, *Journal of Clinical Epidemiology*) audited 3,556 biomedical papers that included a data availability statement claiming the data were available upon reasonable request. They then systematically attempted to obtain the data by contacting authors. The results were stark: 79% of authors either did not respond, refused to share, or shared data that was incomplete or unusable. Less than 7% of the claimed-available datasets were actually accessible.

A parallel assessment using F-UJI on the small fraction of datasets that were deposited in repositories found that even deposited datasets frequently failed key FAIR sub-principles: only 32% had persistent identifiers, 41% had machine-readable metadata, and fewer than 20% used a recognized vocabulary for describing their variables. The datasets existed but were not practically reusable.

This study illustrates the two-layer failure mode that FAIR assessment reveals: the first layer is non-deposition (data never shared despite stated availability), detectable through audit. The second layer is inadequate FAIR compliance in deposited data (data technically available but practically unusable), detectable through automated assessment tools. Both layers contribute to the broader data stewardship failure that FAIR assessment is designed to diagnose.

A second application comes from the NIH data sharing mandate implemented in January 2023, which requires that all NIH-funded research generating scientific data include a data management and sharing plan. The NIH Office of Data Science Strategy uses F-UJI-like assessment frameworks to evaluate compliance with the mandate, identifying which funded projects are producing FAIR-compliant data deposits and which are meeting the letter but not the spirit of the requirement.

5. Requirements & Applicability

FAIR assessment tools have very different requirements from the statistical detection tools in this guide. They evaluate infrastructure and metadata quality rather than statistical reporting, and they operate on data repositories rather than published papers.

A dataset landing page or persistent identifier	F-UJI and the FAIR Evaluator require a URL or DOI pointing to the dataset's landing page or metadata record. Datasets without any online presence cannot be assessed. This is itself a FAIR failure (sub-principle F1: data must have a globally unique and persistent identifier).
Publicly accessible metadata	Automated tools assess metadata that is publicly retrievable via standard web protocols. Datasets in restricted repositories, or whose metadata requires authentication to access, will score lower than their actual compliance warrants. For restricted data, manual assessment using a rubric is the alternative.
No raw data access required	FAIR assessment tools evaluate metadata and structural properties, not the data content itself. The data files need not be downloaded. This makes assessment scalable to large repository collections.

Repository or field scope definition	FAIR assessment is most informative when applied systematically to a defined collection: all datasets deposited in support of papers published in a specific journal, all datasets funded by a specific grant program, or all data in a disciplinary repository. Single-dataset assessments are less interpretable without a baseline comparison.
Domain context for interpretation	FAIR sub-principle scores require domain knowledge to interpret. A dataset in a field with weak community metadata standards may score lower than a dataset in a field with mature ontologies, even if the first is more carefully curated. Comparison across fields without accounting for infrastructure maturity can be misleading.

6. FAIR Assessment as a Research Integrity Signal

FAIR compliance is typically framed as a data management goal rather than an integrity tool. But systematic FAIR assessment reveals patterns that are directly relevant to research integrity at the field level. Several specific signals are worth highlighting for graduate seminars and meta-research contexts.

Data availability statement audits

The gap between stated and actual data availability — documented by Gabelica et al. (2022) and Vines et al. (2014) — is itself a form of misrepresentation. Papers that claim data are available when they are not are reporting false information in the methods or transparency sections. Systematic audit of data availability statements, combined with F-UJI assessment of actually deposited datasets, provides a field-level measure of transparency integrity.

Selective data retention

The FAIR principle that metadata should remain accessible even after data deletion (sub-principle A2) exists precisely because selective deletion of data is a form of evidence manipulation. When a dataset is deleted after publication — particularly when the deletion occurs following concerns about the underlying research — the metadata record provides at least a forensic trace of what existed. Repositories that implement A2 correctly preserve this audit trail; those that do not enable clean evidence destruction.

Provenance as a chain of custody

The FAIR reusability principle requires that data be associated with their provenance — the record of how data were collected, processed, and transformed from raw measurements to the form in which they are deposited. Provenance metadata is the data equivalent of a laboratory notebook: it documents the chain of custody from observation to archive. Datasets with complete provenance metadata can be audited for consistency with the methods section of the associated paper; datasets without provenance cannot.

Compliance rates as field-level integrity indicators

When FAIR assessment is applied systematically across a field's data deposits, the resulting compliance statistics become an integrity indicator. A field in which fewer than 20% of datasets have persistent identifiers, or in which 79% of stated-available datasets are actually inaccessible, has a structural integrity problem that no amount of statistical sophistication in individual papers can compensate for. These field-level rates are now being tracked by funders, publishers, and research assessment frameworks as components of responsible research practice evaluation.

7. Limitations

- FAIR compliance measures infrastructure quality, not scientific quality. A meticulously described dataset of fabricated measurements is FAIR-compliant. High FAIR scores are a necessary but not sufficient condition for data integrity.
- Automated assessment tools evaluate metadata properties, not the accuracy or completeness of the metadata itself. A dataset can have a DOI, use a recognized vocabulary, and carry an open license, while containing inaccurate variable descriptions or omitting crucial methodological details. FAIR assessment cannot catch lies told in metadata.

- FAIR sub-principle scores are not fully standardized across tools. F-UJI, FAIRshake, and the FAIR Evaluator use different scoring rules for some criteria, making cross-tool comparisons unreliable. When reporting FAIR compliance results, always specify which tool and which version was used.
- Many disciplines lack mature community metadata standards — the vocabulary and ontology infrastructure that FAIR interoperability and reusability require. In such fields, even well-intentioned data depositors will score poorly on these sub-principles through no fault of their own. FAIR assessment in immature data ecosystems measures the infrastructure gap as much as individual compliance.
- FAIR does not address data quality, completeness, or representativeness. A dataset can be perfectly FAIR-compliant and still be a biased sample, collected with a flawed instrument, or analyzed with inappropriate methods. The FAIR framework is about access and reusability, not scientific validity.

8. How to Apply

Assessing an individual dataset (F-UJI)

Navigate to f-uji.net and enter the dataset's DOI or landing page URL. The tool returns a scored report within seconds. Review the sub-principle scores, noting which categories (F, A, I, R) show low maturity. The most common failures in published research datasets are: no persistent identifier (F1), no structured machine-readable metadata (F2), restrictive or absent license (R1.1), and no provenance information (R1.2).

Batch assessment of a repository collection

Use the F-UJI REST API (documented at f-uji.net/api) to programmatically assess a collection of datasets. Collect DOIs from a bibliometric database or journal data availability statements, submit each to the API, and aggregate the JSON results. The resulting dataset of sub-principle scores can then be analyzed to produce field-level compliance statistics comparable to those reported by Gabelica et al. (2022).

Auditing data availability statements

For a systematic audit: (1) collect all data availability statements from a sample of papers in the target journal or field; (2) classify each statement (data in repository / available on request / not available / no statement); (3) for statements claiming repository deposit, extract the DOI or URL and run F-UJI; (4) for statements claiming availability on request, contact authors following the protocol used by Gabelica et al. Document response rates, refusal rates, and the FAIR compliance of actually obtained datasets. The gap between stated and actual availability, and between deposited and FAIR-compliant datasets, are the two primary integrity signals.

9. Related Tools

DataCite	Global DOI registration agency for research data. DataCite DOIs are the most widely used persistent identifiers for datasets (FAIR sub-principle F1). The DataCite Metadata Schema specifies the structured metadata required for registration, covering many F and R sub-principles automatically for registered datasets.
ORCID	Persistent researcher identifier that links publications, datasets, and other outputs to a verified scholarly identity. Contributes to FAIR provenance (R1.2) by unambiguously attributing data to specific researchers across institutions and career changes.
Zenodo / Figshare / Dryad	Open data repositories that implement FAIR principles structurally: automatic DOI assignment (F1), open HTTP access (A1), Dublin Core and DataCite metadata standards (I1, R1), and Creative Commons licensing (R1.1). Depositing in these repositories confers substantial FAIR compliance without additional metadata effort.
Research Object Crate (RO-Crate)	A lightweight standard for packaging research data with structured metadata using Schema.org vocabularies. RO-Crate directly implements FAIR interoperability (I1–I3) and provenance (R1.2) requirements. Increasingly adopted in workflow management systems and computational reproducibility pipelines.
Registered Reports	Preregistration format that commits to data sharing before data collection. Journals offering Registered Reports typically require FAIR-compliant data deposits as a condition of publication — creating a structural link between study pre-registration and data accessibility that addresses the data availability gap.

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Looking Ahead: Scaling Open Science

COMMENTARY

Convergence: Detection Tools, Ontologies & Scalability

At a glance: The statistical detection tools described in this guide share a common bottleneck: they operate on unstructured, field-specific, inconsistently reported data. Formal ontologies — machine-readable frameworks that define concepts and their relationships — offer a route to making these tools systematic, interoperable, and scalable. This chapter examines where ontologies are already doing this work, what they require, and where the gaps remain.

1. The Scalability Problem

Each tool described in this guide — GRIM, p-curve, PET-PEESE, bibliometric anomaly detection — was designed by researchers working in a specific domain, solving a specific detection problem with whatever data happened to be available. Each requires the analyst to manually locate the right studies, extract the right statistics, verify that the assumptions are met, and interpret results against field-specific norms. This is scholarly work, not automated screening.

The practical consequence is that despite two decades of methodological development, the detection tools described in this guide remain primarily the instruments of specialists. A meta-scientist running a p-curve analysis on a literature of one hundred studies is doing weeks of work. A journal editor wanting to screen submitted meta-analyses for publication bias has no automated pipeline. A funding agency trying to assess the integrity of a research program has no dashboard. The tools exist; the infrastructure to deploy them at scale does not.

This is not primarily a computational problem — the algorithms are fast once the inputs are in the right format. It is a data representation problem. The inputs are locked in inconsistently formatted PDFs, described with field-specific terminology that varies across journals and decades, and linked to study metadata that is either missing or stored in incompatible systems. Solving the scalability problem means solving the data representation problem first. This is precisely what ontologies are designed to do.

2. What Ontologies Are and Why They Matter Here

An ontology, in the knowledge representation sense, is a formal specification of a domain: a controlled vocabulary of concepts, a taxonomy of how those concepts relate hierarchically, and a set of formal axioms that constrain how instances of those concepts can be combined. The word is borrowed from philosophy (ontology = the study of what exists) but in computer science it refers specifically to machine-readable knowledge structures that support automated reasoning.

Ontologies differ from simple controlled vocabularies (like MeSH or thesauri) in two important ways. First, they express relationships between concepts, not just a list of terms — an ontology can encode that 'randomized controlled trial' is a subtype of 'experimental study', that 'sample size' is a property of a 'study design', and that 'Cohen's d' is a type of 'standardized mean difference effect size'. Second, they support inference: a software agent that knows an ontology can conclude things that are not explicitly stated, by following the logical relationships.

For open science detection tools, three capabilities of ontologies are especially relevant:

- **Interoperability:** if two databases both use the same ontology to describe their studies, a detection tool can query across both without manual harmonization. The FAIR principles (Findable, Accessible, Interoperable, Reusable) depend heavily on ontological annotation to achieve the 'I' — interoperability without human translators.
- **Automated extraction:** an ontologically annotated paper allows software to locate the sample size, the effect size metric, the hypothesis being tested, and the population being studied without reading the paper. This is the prerequisite for automated p-curve construction, bibliometric analysis, and replication matching.
- **Consistency checking:** an ontology-aware system can flag papers where reported values are inconsistent with the stated study design — a form of GRIM-like checking that generalizes beyond integer means to any formally constrained reporting convention.

3. Where Ontologies Are Already in Use

Biomedicine: the OBO Foundry and its ecosystem

The most mature application of ontologies to research practice is in biomedicine, where the Open Biological and Biomedical Ontology (OBO) Foundry coordinates a suite of interoperable ontologies covering organisms, anatomical structures, diseases, phenotypes, chemical entities, and experimental methods. Several of these are directly relevant to research integrity and open science detection:

Ontology for Biomedical Investigations (OBI)	Defines experimental designs, assays, data transformations, and statistical analyses. An OBI-annotated study explicitly encodes its design type, sample size, measurement type, and analysis method — exactly the inputs needed for GRIM, p-curve, and requirements checking.
Human Phenotype Ontology (HPO)	Used in clinical genomics to standardize phenotype descriptions across databases. Enables automated matching of patient cohorts across studies — a prerequisite for systematic replication and cross-study effect size comparison.
Clinical Data Interchange Standards Consortium (CDISC)	Not an ontology per se but a metadata standard widely used in pharmaceutical trials. CDISC-formatted trial data can be queried programmatically for sample sizes, primary endpoints, and outcomes — enabling automated CONSORT compliance checking and outcome switching detection.
Gene Ontology (GO)	The original large-scale biological ontology. Demonstrates at scale what becomes possible when a community commits to a shared vocabulary: cross-database queries, automated annotation, and computational meta-analysis that would be impossible with free-text descriptions.
Research Data Alliance (RDA) vocabularies	Community-maintained metadata vocabularies for data types, repositories, and access conditions. Supports FAIR compliance assessment by providing the controlled terms needed to evaluate whether metadata is machine-readable and interoperable.

Precision medicine: ontologies as infrastructure for evidence synthesis

Precision medicine — the program of matching treatments to patients based on molecular, phenotypic, and contextual profiles — depends on exactly the kind of structured, interoperable evidence base that ontologies enable. The Global Alliance for Genomics and Health (GA4GH) has developed phenopackets, a structured data format for clinical observations that uses HPO, OMIM, and NCIT controlled terms to describe patients in a way that is both human-readable and machine-queryable.

The relevance to open science detection is direct. When clinical trial results are reported using GA4GH phenopackets and CDISC data models, the primary outcomes, patient characteristics, and effect sizes are machine-readable at the point of publication. A system that can read phenopackets can automatically extract the inputs needed to run a p-curve analysis across a therapeutic area, check for outcome switching against a trial registry record, or compute a TES excess significance test — without a human analyst ever reading the full paper text.

The All of Us Research Program and the UK Biobank both use structured phenotype ontologies to annotate their cohorts. This has enabled automated replication studies — computational searches for studies that used comparable populations and endpoints — that would be infeasible with free-text descriptions. The same infrastructure is beginning to be used to flag studies that claim replication of UK Biobank findings but use incompatible phenotype definitions, a subtle form of the outcome-switching problem.

Social and behavioral sciences: emerging standards

The social and behavioral sciences lag substantially behind biomedicine in ontological infrastructure, but several initiatives are building the foundations. The Variable and Value Label Ontology (VVLO) and the Data Documentation Initiative (DDI) provide structured formats for survey instruments and their response scales — directly relevant to GRIM applicability, since the integer scale structure that GRIM requires is machine-readable in a DDI-annotated dataset. The Cognitive Atlas provides a formal ontology of psychological constructs and tasks, enabling automated identification of which studies are estimating comparable effects — a prerequisite for valid p-curve analysis across a literature.

The Psychological Science Accelerator and similar big-team science networks have begun requiring structured metadata at the point of preregistration, using templates that enforce controlled vocabulary for

design type, manipulation type, and outcome measure. When preregistrations use controlled terms rather than free text, automated comparison of preregistered and published outcomes becomes computationally feasible rather than requiring human coding.

4. Core Features of Ontologies Useful for Detection

Not every ontology enables detection tool scalability. Several structural features distinguish those that do from those that merely provide a vocabulary.

Feature	Detection Capability	Example
Feature	What it enables	Example
Formal axioms	Automated consistency checking — flagging impossible combinations of reported values	OBI axioms that constrain which statistical tests are valid for which study designs
Persistent URIs	Stable cross-database linking — the same concept has the same identifier across journals, databases, and decades	OBO PURL system: each term has a permanent URL that never changes
Modular design	Reuse across domains — a study design ontology can import measurement terms from a separate measurement ontology without duplication	OBO Foundry interoperability principle: modules import rather than redefine
Mappings to existing standards	Integration with legacy data — ontology terms are mapped to MeSH, ICD, SNOMED entries so historical literature can be retrospectively annotated	OMOP CDM mappings enable retrospective EHR analysis using ontology queries
Version control	Reproducibility of annotations — specifying which version of an ontology was used at the time of annotation ensures the annotation remains interpretable as terms evolve	OWL format with versioning IRI; BioPortal version history
Community governance	Long-term maintenance and domain validity — concepts are added and revised by domain experts through transparent processes, not unilaterally	OBO Foundry review process; Gene Ontology Consortium

5. The Convergence Opportunity

The detection tools described in this guide and the ontological infrastructure emerging from precision medicine and open science programs are on a collision course — in a productive sense. Each addresses a problem the other cannot solve alone.

The detection tools solve the statistical problem: given a set of study results, how do we diagnose p-hacking, publication bias, analytical fragility, or data fabrication? But they are bottlenecked by the data extraction problem — getting the right numbers from unstructured papers at scale.

The ontologies solve the data representation problem: given a formal description of a study's design, measurements, and results, the relevant inputs for detection tools can be extracted automatically. But ontologies, by themselves, do not perform any detection — they are passive knowledge structures. The detection logic has to be implemented separately.

The convergence opportunity is to connect these two layers: ontologically annotated study metadata feeding automatically into detection tool pipelines, producing systematic, field-wide integrity assessments rather than individual case-by-case analyses. Several specific integration points are already technically feasible:

“The question is no longer whether we have the tools to detect systematic bias at scale. The question is whether the knowledge infrastructure exists to feed those tools without an army of human annotators. Ontologies are the answer to that question — but only if they are adopted upstream, at the point of data collection and reporting, not retrofitted onto the published literature.”

— Synthesized from Wilkinson et al. (2016) FAIR Principles and Ioannidis (2014) on meta-research infrastructure

Integration point 1: Automated p-curve and p-uniform inputs

If published papers annotate their primary outcome tests using a structured reporting ontology (specifying the test type, the test statistic, the degrees of freedom, and the hypothesis direction in machine-readable form), a p-curve or p-uniform analysis over an entire literature can be constructed algorithmically. The Semantic Publishing and Referencing Ontologies (SPAR) — particularly the Statistical Methods Ontology (STATO) — provide exactly this vocabulary. STATO defines t-tests, F-tests, chi-squared tests, and their parameters in OWL, making them queryable across any database that uses STATO annotations.

The Linked Open Data publishing platform for trials, currently piloted by several European research funders, uses STATO annotations in registered and published trials. A proof-of-concept p-curve across all CONSORT-compliant registered trials in a therapeutic area can be computed in minutes once the data is in this format — a task that currently requires months of manual extraction.

Integration point 2: Cross-literature replication matching

One of the most significant open problems in the reproducibility movement is identifying which studies are genuine replications of each other. The Cognitive Atlas and OBI together can support computational replication matching: two studies that share the same Cognitive Atlas task term, the same OBI measurement type, and the same OBI statistical test are strong candidates for replication pairs. This enables automated construction of replication networks — a prerequisite for running z-curve 2.0 or meta-analytic correction methods across a domain rather than a manually curated literature.

The Institute for Replication (i4rep) has begun developing a structured replication registry that uses controlled vocabulary for the original study and replication relationship. As this scales, it becomes possible to automatically compute field-wide replication rates, calibrate TES power estimates against actual replication outcomes, and identify which research programs have systematically low replication rates without requiring a human expert to read each study pair.

Integration point 3: Outcome switching detection at submission

The most promising near-term application of ontology-enabled detection is at the journal submission stage. If a journal requires structured reporting of primary and secondary outcomes in machine-readable form (using CDISC or a compatible vocabulary), and if the trial registry record is also in structured form, the discrepancy between registered and published outcomes can be detected automatically at the moment of submission — before peer review, before publication, and before the paper enters the meta-analytic record.

EQUATOR Network's Registered Report template, combined with the SPIRIT checklist (which has a machine-readable XML implementation), creates the data model for this detection. Several publishers including F1000Research and PLOS Medicine have implemented structured outcome reporting fields in their submission systems. The missing piece is requiring machine-readable registry linkage at submission rather than as a recommendation.

6. Challenges and Open Problems

- Adoption is the fundamental barrier. An ontology that most researchers do not use provides no benefit. Biomedical databases achieved adoption through regulatory requirement (FDA and EMA mandate CDISC for trial submissions) and funder mandate (NIH requires FAIR data management plans). Social and behavioral sciences lack equivalent regulatory leverage, and voluntary adoption has been slow.
- Retrospective annotation is expensive. The existing published literature — millions of papers — is not ontologically annotated and cannot be automatically annotated without error. Large language models (GPT-4, Llama, domain-specific models) are beginning to change this calculus: they can extract

structured metadata from free text at scale, albeit imperfectly. Hybrid workflows combining LLM extraction with ontology-based validation are a promising near-term approach.

- Ontologies age. Concepts evolve, measurement instruments are updated, statistical practices change. An ontology annotation made in 2010 using a term that has since been deprecated is difficult to interpret. The versioning and provenance requirements for useful ontologies are substantially more demanding than for simple metadata schemas.
- Gaming is possible. If journals require structured outcome reporting for detection-tool screening, researchers who wish to misrepresent their analyses could provide structured metadata that does not match the actual analysis. This is a real concern, but it raises the cost and visibility of misrepresentation — structured lies are easier to detect than free-text misrepresentation, because they can be automatically checked for internal consistency and cross-validated against registered protocols.
- Field-specificity limits generalizability. An ontology developed for clinical trials in oncology is not directly applicable to social psychology experiments or economics field experiments. The OBO Foundry's modular approach is the right architectural solution — shared upper-level ontologies with domain-specific extensions — but building those extensions requires sustained community investment in each field.

7. Implications for the Detection Tools in This Guide

Reading the tools in this guide through the lens of ontological infrastructure reveals a gradient of scalability potential.

GRIM / SPRITE / statcheck	Highest near-term scalability. These tools require only summary statistics (mean, SD, N, test statistic). Structured reporting of these values in machine-readable form is already recommended by APA and enforced by some journals. An OBI-annotated paper provides all required inputs automatically.
p-curve / p-uniform	High potential with structured statistical reporting. STATO annotations provide the test statistics and df required. The bottleneck is focal test identification — determining which of many reported tests is the primary outcome — which requires ontological encoding of the study's hypothesis structure, available in OBI but rarely populated.
Funnel plot / PET-PEESE / TES	Moderate potential. Meta-analytic tools require effect sizes and standard errors aggregated across studies. If individual study results use structured effect size reporting (as recommended by the APA Task Force on Statistical Reporting and encoded in STATO), meta-analytic inputs can be assembled automatically from literature search results.
Multiverse / Specification Curve	Lower near-term scalability. These tools require access to raw data and analysis code, not just reported statistics. FAIR data principles and code availability mandates are preconditions, but the additional requirement for structured description of the analytic choice space has no current ontological standard.
Bibliometric Anomaly Detection	Already partially automated. Bibliometric databases (OpenAlex, Semantic Scholar) provide the citation graph and authorship metadata required. The main ontological gap is in author disambiguation (ORCID partially solves this) and topic classification (which determines what counts as an anomalous citation pattern relative to the field).
Outcome switching audit	Highest leverage from ontological infrastructure. The comparison of registered to published outcomes is fundamentally a data matching problem — it requires that both the registry record and the paper use the same controlled vocabulary for outcomes. This is solvable now with existing ontologies; the barrier is policy, not technology.

8. A Path Forward

The convergence of detection tools and ontological infrastructure suggests a practical agenda organized around three time horizons.

Near term (1–3 years): require structured statistical reporting

The highest-leverage immediate intervention is mandating structured, machine-readable statistical reporting at the point of journal submission. This means requiring test statistics, degrees of freedom, effect sizes, and sample sizes in a standardized format — not just as prose in a results section but as structured metadata that can be extracted without natural language processing. Several journals already do this in medical fields; extending the requirement to social and behavioral sciences would immediately enable automated GRIM checking, statcheck screening, and structured p-curve construction across any literature covered by these journals.

Medium term (3–7 years): ontology-enabled replication infrastructure

Building a cross-domain replication registry that uses controlled vocabulary for study designs, populations, manipulations, and outcomes would enable systematic computation of replication rates, z-curve analyses, and TES tests across fields rather than within manually curated literatures. The Institute for Replication, the Psychological Science Accelerator, and equivalents in economics and medical science provide the institutional nuclei. The technical infrastructure — based on existing OBO Foundry and SPAR ontologies — is largely available; what is needed is community agreement on a minimal shared metadata standard for replication registration.

Long term (7+ years): integrated research integrity information systems

The long-term vision is a research information ecosystem in which integrity signals — bibliometric anomalies, p-value distribution warnings, outcome switching flags, GRIM inconsistencies — are computed continuously and attached to papers and research programs as persistent metadata, in the same way that citation counts and altmetrics are currently attached. This is technically feasible given the infrastructure being built in precision medicine and the detection tools now available. The barriers are governance (who manages the integrity metadata?), liability (what are the consequences of a false positive flag?), and incentives (who pays for the annotation infrastructure?). None of these barriers is insurmountable, but none is primarily a technical problem.

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